

INSTABILITIES AND BIFURCATION STRUCTURE IN A MECHANOCHEMICAL MODEL OF PATTERN FORMATION

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ABSTRACT. We analyze the stability and bifurcation structure of steady states in a mechanochemical model of pattern formation in regenerating tissue spheroids. The model couples morphogen dynamics with tissue mechanics via a positive feedback loop: mechanical stretching enhances morphogen production, while morphogen concentration modulates tissue elasticity. Global strain conservation implements a nonlocal inhibitory effect, realizing a mechanochemical variant of the local activation–long-range inhibition mechanism. For exponential elasticity–morphogen coupling, the system admits a variational formulation. We prove existence of nonconstant steady states for small diffusion and uniqueness of the homogeneous state for large diffusion. Linear stability analysis shows that only unimodal patterns are stable, while multimodal solutions are unstable. Bifurcation analysis reveals subcritical and supercritical pitchforks, with fold bifurcations generating bistable regimes. Our results demonstrate that mechanochemical feedback provides a robust mechanism for single-peaked pattern formation without requiring a second diffusible inhibitor.

1. INTRODUCTION

Self-organized pattern formation refers to the spontaneous emergence of spatially heterogeneous states from homogeneous equilibria in spatially extended systems governed by local interactions. In biological modeling, such phenomena are commonly described by coupled partial differential equations combining nonlinear reaction–diffusion dynamics with mechanical feedback, leading to symmetry breaking and spatial organization in developing or regenerating tissues. Despite extensive modeling efforts, a rigorous mathematical understanding of how nonlinear chemical kinetics interacting with mechanical couplings give rise to robust pattern selection remains incomplete [15, 31]. In particular, nonlocal mechanical constraints introduce qualitative changes in stability properties, mode selection, and bifurcation structure, posing significant analytical challenges for the theory of pattern formation.

Since Turing’s seminal proposal that diffusion-driven instabilities can generate spatial patterns [36], reaction-diffusion models have become a central theoretical framework for biological pattern formation [20]. Canonical examples such as the Gierer-Meinhardt model [14] formalized the local activation–long-range inhibition (LALI) principle, showing how nonlinear interactions between an autocatalytic activator and a spatially extended inhibitory signal can

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give rise to stable patterns. In these classical Turing-type systems [25, 32, 38], pattern formation relies on at least two interacting morphogens with sufficiently distinct effective ranges, typically realized through differential diffusion. Despite their conceptual success, identifying concrete molecular implementations of Turing systems *in vivo* has remained notoriously difficult [23]. In particular, the nature of the long-range inhibitory signal is often unclear, raising the possibility that mechanisms beyond simple molecular diffusion may underlie observed patterns. Motivated by this, kernel-based formulations of pattern formation have been proposed, in which spatial interactions are encoded directly through nonlocal activation–inhibition kernels rather than diffusion operators [10, 19]. Such models can reproduce the full repertoire of classical two-dimensional patterns, including spots, stripes, and networks, without explicitly invoking diffusive transport. More generally, it was demonstrated that reaction–diffusion systems with arbitrarily many components can be reduced to integro-differential descriptions characterized by effective kernels [12], implying that distinct molecular mechanisms may generate identical spatial patterns if they share the same interaction structure. While powerful, this abstraction also lacks direct biological interpretation, as the cellular or molecular origins of the effective kernels are often not directly identifiable.

Growing experimental evidence indicates that mechanical forces are integral to biological organization across scales, challenging purely chemical views of pattern formation [35]. This has motivated mechanochemical frameworks in which tissue mechanics and biochemical signaling are dynamically coupled. These questions are often investigated using biologically motivated mathematical models, with patterning in the freshwater polyp *Hydra* providing a prominent example [30, 39]. Existing *Hydra* models range from classical reaction-diffusion descriptions of activation–inhibition dynamics [14, 24] to more recent formulations based on receptor-mediated interactions coupled to intracellular signaling [27, 42]. However, purely biochemical models appear insufficient to account for observed patterning behavior in *Hydra*. Both experimental and theoretical studies indicate that mechanical effects can act as effective long-range interactions and generate robust spatial organization [13, 17, 26]. Mechanochemical models naturally give rise to feedback loops between deformation and signaling [41], suggesting that mechanical coupling can replace the classical diffusible inhibitor and provide an alternative realization of the LALI mechanism [5, 13, 40].

Several theoretical frameworks have incorporated mechanical effects into pattern formation models, ranging from simple mechanochemical feedbacks that generate coupled chemical–mechanical patterns [6, 26] to more elaborate continuum descriptions, such as biphasic poroelastic tissue models, in which mechanically induced flows enable pattern formation even for minimal biochemical kinetics [34]. Despite this progress, a systematic mathematical understanding of pattern selection and stability in mechanochemical systems remains incomplete. In contrast to classical Turing models, where mode selection and stability are well characterized, the influence of mechanical coupling on pattern multiplicity, robustness, and minimal pattern-forming requirements is still poorly understood.

In this work, we address these gaps by analyzing the stability and bifurcation structure of stationary solutions in a new mechanochemical model of symmetry breaking in *Hydra* aggregates, which was recently validated against experimental observations [41]. The model couples morphogen dynamics to tissue mechanics via a positive feedback mechanism: mechanical stretching enhances morphogen production, while morphogen concentration modulates tissue elasticity. Together with a global strain conservation constraint, this coupling yields a mechanochemical variant of the LALI paradigm that does not require a second diffusible inhibitor. We focus on an exponential elasticity–morphogen coupling, which endows the system with a variational structure amenable to rigorous analysis.

By deriving a reduced one-dimensional model that preserves the essential mechanochemical feedback while simplifying the geometric setting (Section 3, Appendix A), we obtain, for the

first time, a framework amenable to rigorous analytical investigation of the dynamics. We establish the existence of spatially heterogeneous steady states for sufficiently small diffusion coefficients and prove uniqueness of the homogeneous equilibrium for large diffusion, deriving explicit bounds on the corresponding critical values by variational methods (Section 4). This analysis explains the mechanism of symmetry breaking, demonstrating how patterned states bifurcate from the homogeneous configuration. We then investigate the pattern selection problem (Section 5) and show that only unimodal stationary patterns are linearly stable, while all multimodal solutions are unstable, revealing a fundamental distinction from classical Turing systems and curvature-based mechanochemical models [31, 38]. Finally, we identify both subcritical and supercritical pitchfork bifurcations and reveal additional fold bifurcations responsible for the emergence of bistable regimes (Section 6). Throughout, the theoretical results are complemented by numerical simulations, elucidating the interplay between nonlocality, mechanical feedback, and nonlinear kinetics, and providing a mathematically grounded framework for experimental exploration.

Notation. We use the following notation: $\mathbb{T}$ is a one dimensional torus of size one, i.e., the unit interval $[0, 1]$ in which we identify the endpoints with each another. Here $L^2(\mathbb{T})$ denotes the set of square integrable functions on the circle defined via Fourier series

$$L^2(\mathbb{T}) = \left\{ u = \sum_{n=-\infty}^{\infty} \hat{u}(n) e^{2\pi i n x} : \sum_{n=-\infty}^{\infty} |\hat{u}(n)|^2 < \infty \right\}$$

Similarly we denote Sobolev space

$$W^{k,2}(\mathbb{T}) = \left\{ u \in L^2(\mathbb{T}) : \sum_{n=-\infty}^{\infty} (1 + n^2)^k |\hat{u}(n)|^2 < \infty \right\}.$$

Constants are always denoted by a letter C even if they vary from line to line.

2. MAIN RESULTS

We start by formulating main theoretical results of this paper. We focus the mechanochemical pattern formation model describing the concentration of the diffusive signaling molecules $u := u(x, t)$, given by

$$(2.1) \quad \partial_t u(x, t) = D \partial_{xx} u(x, t) - u(x, t) + \kappa \frac{e^{u(x, t)}}{\int_0^1 e^{u(y, t)} dy}, \quad x \in [0, 1], \quad t \in [0, \infty),$$

with the coefficients $D > 0$ and $\kappa > 0$, supplemented with the periodic boundary conditions $u(0, t) = u(1, t)$ and $\partial_x u(0, t) = \partial_x u(1, t)$ for all $t \geq 0$. This model arises from a one-dimensional formulation of a hyperelastic material governed by the Saint-Venant–Kirchhoff constitutive law. A detailed derivation is provided in Section 3 and Appendix A. Numerical simulations of the model (Figure 2.1) demonstrate its capacity to produce stable unimodal patterns.

Global existence and uniqueness are classical results, established by standard arguments (see, e.g., [28]); we recall them here for completeness,

Proposition 2.1 (Global existence and uniqueness). *For any initial datum $u_0 \in W^{1,2}(\mathbb{T})$ problem (2.1) admits a unique global in time solution $u = u(x, t)$ satisfying*

$$u \in C([0, \infty), W^{1,2}(\mathbb{T})), \quad u \in L^2([0, \infty), L^2(\mathbb{T})), \quad u \in L^2((0, T), W^{2,2}(\mathbb{T})) \quad \text{for each } T > 0.$$

A complete proof is provided in Appendix for the reader's convenience.

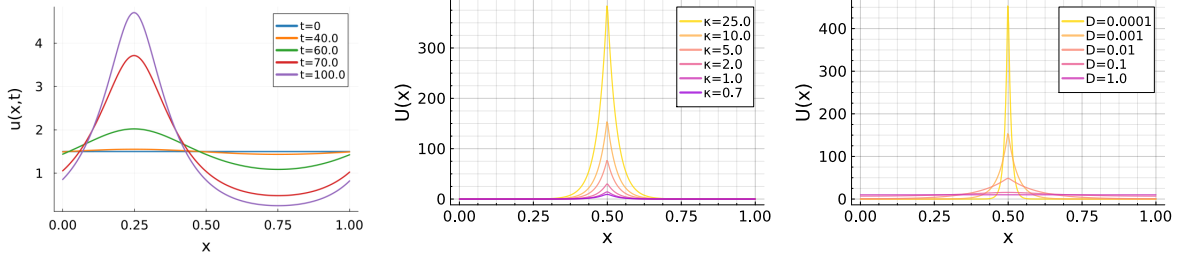


FIGURE 2.1. Numerical simulations of model (2.1). Left panel: Emergence of a uni-modal pattern from a small perturbation of the constant stationary solution $U = \kappa$, (example for $D = 0.01$, $\kappa = 1.5$). Middle panel: Nonconstant stationary solutions for fixed diffusion $D = 10^{-3}$ and varying κ . Right panel: Nonconstant stationary solutions for fixed $\kappa = 3.0$ and varying D . We observe that smaller diffusion leads to more concentrated patterns.

We now characterize stationary solutions $u(x, t) = U(x)$ of (2.1) (see Figure 2.1), which satisfy

$$(2.2) \quad 0 = D U_{xx} - U + \kappa \frac{e^U}{\int_0^1 e^U dy},$$

supplemented with the periodic boundary conditions $U(0) = U(1)$ and $U_x(0) = U_x(1)$.

Remark 2.2 (Mass conservation of stationary structures). Integration of (2.1) over $\mathbb{T}$ yields the mass evolution equation

$$\frac{d}{dt} \int_0^1 u dx = \kappa - \int_0^1 u dx,$$

implying $\int_{\mathbb{T}} u dx \rightarrow \kappa$ as $t \rightarrow \infty$. Thus, any stationary solution $U(x)$ satisfies $\int_{\mathbb{T}} U(x) dx = \kappa$.

Theorem 2.3 (Existence of stationary solutions). *For each $\kappa > 0$ there exist two constants $D_{\min}(\kappa)$ and $D_{\max}(\kappa)$ satisfying*

$$0 < D_{\min}(\kappa) < D_{\max}(\kappa),$$

such that:

- If $0 < D < D_{\min}$, then problem (2.2) admits a nonconstant solution, which is Lyapunov stable with respect to L^2 -norm for perturbations in $W^{1,2}(\mathbb{T})$.
- If $D > D_{\max}$, the constant function $\bar{U} \equiv \kappa$ is the unique solution to equation (2.2) and globally asymptotically stable with respect to L^2 -norm for perturbations in $W^{1,2}(\mathbb{T})$.

The proof, which employs variational methods, is presented in Section 4. We emphasize that the analytically derived bounds $D_{\min}$ and $D_{\max}$ are not optimal, as they arise from relatively coarse estimates; their refinement by numerical simulations is presented at the end of the section.

We classify nonconstant stationary solutions according to their spatial periodicity: an m -modal solution has m peaks per period. In particular, solutions with $m = 1$ are called unimodal, while those with $m = 2, 3, \dots$ are multimodal. By rescaling $x \mapsto mx$, a unimodal solution U of equation (2.2) yields an m -modal solution $\tilde{U}(x) = U(mx - \lfloor mx \rfloor)$ of

$$(2.3) \quad 0 = \frac{D}{m^2} \tilde{U}_{xx} - \tilde{U} + \kappa \frac{e^{\tilde{U}}}{\int_0^1 e^{\tilde{U}} dy},$$

with effective diffusivity D/m^2 .

To investigate the stability properties of stationary solutions of (2.1), we consider perturbations of the form

$$u(x, t) = U(x) + e^{\lambda t} \varphi(x),$$

where U is a stationary solution. Substituting this ansatz into (2.1) leads to the following nonlocal eigenvalue problem:

$$D\varphi_{xx} + \left(\kappa \frac{e^U}{\int_0^1 e^U dy} - 1 - \nu \right) \varphi = \kappa \frac{e^U}{\left(\int_0^1 e^U dy \right)^2} \int_0^1 e^U \varphi dy.$$

We denote by ν an eigenvalue of the nonlocal problem and compare it below with the eigenvalues λ of the corresponding local problem. The following results describe the linear stability of nonconstant stationary solutions.

Theorem 2.4 (Type of stationary solutions). *The stationary solutions are classified as follows:*

- The stable nonconstant solution obtained in Theorem 2.3 is unimodal.
- All multimodal solutions (m -modal with $m \geq 2$) are linearly and nonlinearly unstable in $L^2(\mathbb{T})$.

The proof, based on the gradient flow structure of (2.1) and Sturm–Liouville theory, is presented in detail in Section 5.

Corollary 2.5 (Existence of multimodal solutions). *Combining Theorems 2.3 and 2.4 with equation (2.3), we obtain:*

- If $D < D_{\min}/m^2$, then an m -modal solution to (2.2) exists.
- If $D > D_{\max}/m^2$, then no m -modal weak solutions exist.

This result highlights a fundamental principle: stable pattern formation in our mechanochemical framework necessarily yields unimodal structures, whereas all multimodal configurations are unstable. This represents a significant difference from classical Turing models, in which the number of peaks corresponds to the unstable eigenmodes and can vary with domain size [29]. Pattern selection in our model also differs from other mechanochemical frameworks that permit multiple, or even arbitrarily many, peaks, see, e.g., [6, 26, 31].

We complete the analysis by characterizing the emergence of patterns through bifurcations from the homogeneous steady state.

Theorem 2.6 (Bifurcation structure). *For each $n \in \mathbb{N}$, define $\kappa_n := 1 + 4\pi^2 n^2 D$. Then the following holds:*

- (1) **Existence.** *At each bifurcation point $(\bar{\Phi}_n, \kappa_n)$, with $\bar{\Phi}_n = \kappa_n$, there exists a smooth local branch of nonconstant solutions*

$$\Gamma_n(s) = (\bar{\varphi}_n(s), \kappa_n(s)),$$

admitting the following asymptotic expansion as $s \rightarrow 0$

$$\bar{\varphi}_n(s) = \kappa_n + s\sqrt{2} \cos(2\pi n x) + \mathcal{O}(s^2), \quad \kappa_n(s) = \kappa_n + C_n s^2 + \mathcal{O}(s^3),$$

where $C_n = \frac{1}{4} - \frac{1}{16Dn^2\pi^2} + 2Dn^2\pi^2$.

- (2) **Type.** *The bifurcation at $(\bar{\Phi}_n, \kappa_n)$ is*
 - *subcritical if $\kappa_n < \frac{3}{2}$, or equivalently $D < \frac{1}{8\pi^2 n^2}$,*
 - *supercritical if $\kappa_n > \frac{3}{2}$, or equivalently $D > \frac{1}{8\pi^2 n^2}$.*

- (3) **Stability.** The branch $\Gamma_n(s)$ is linearly stable near the bifurcation point if and only if $n = 1$ and $\kappa_1 > \frac{3}{2}$. All remaining branches are linearly unstable in a neighborhood of the bifurcation point. Moreover, this local branch admits a global continuation in the sense of Rabinowitz and
- (4) **Turning point.** If $\kappa_1 < \frac{3}{2}$, the branch $\Gamma_1(s)$ undergoes a fold bifurcation at some $\kappa_f \in (0, \kappa_1)$.

Theorem 2.6 summarizes the bifurcation analysis carried out in Section 6. It combines local bifurcation results obtained via the Crandall–Rabinowitz theorem with a global continuation analysis in the sense of Rabinowitz, thereby revealing the presence of fold bifurcations and bistable regimes. Together, these results show that the transition from the homogeneous to the patterned state is mediated by a bifurcation whose direction and stability depend critically on the balance between diffusion and reaction strength. In particular, the bifurcation undergoes a change of type at a critical parameter threshold, a phenomenon also observed in other pattern formation systems [2]. We also identify a mathematical mechanism for the emergence of bistability, corresponding to the coexistence of homogeneous and patterned steady states. Such bistable behavior has been recently reported in *Hydra* regeneration [37].

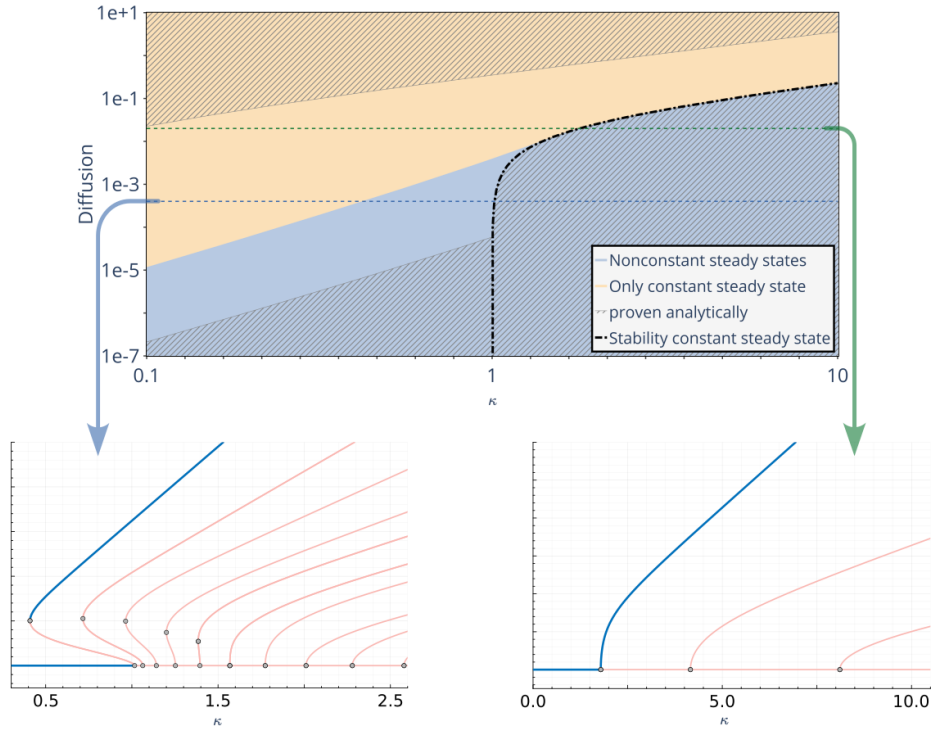


FIGURE 2.2. Parameter space analysis and bifurcation structure. (Top) Parameter space on a logarithmic scale (D, κ) showing regions of numerical convergence to nonconstant (blue) versus constant (yellow) steady states, with gray areas indicating analytically proven results. (Bottom) Bifurcation diagrams demonstrating subcritical regime with bistability (left, $D < 1/(8\pi^2)$) and supercritical regime with direct pattern formation (right, $D > 1/(8\pi^2)$), where blue and red curves denote stable and unstable branches, respectively.

We conclude this section by illustrating our analytical results through numerical computations. The analytical bounds $D_{\min}(\kappa)$ and $D_{\max}(\kappa)$ obtained in Theorem 2.3 are not expected to be sharp, as they arise from relatively coarse estimates. Nevertheless, the numerical results

presented below demonstrate that these bounds correctly capture the qualitative structure of the parameter space. In particular, Figure 2.2 confirms the predicted transition between subcritical and supercritical bifurcation regimes at the critical threshold. Moreover, the numerical simulations illustrate the existence of bistable parameter regions arising from fold bifurcations and thereby support the overall bifurcation framework developed in the analytical part of this work. While Theorem 2.6 provides a complete characterization of bifurcating branches in the neighborhood of each bifurcation point, the global continuation of these branches lies beyond the reach of the analytical techniques employed here. In the subcritical regime $\kappa < 1.5$ the theory guarantees only that the primary branch $\Gamma_1(s)$ possesses at least one turning point. Our numerical analysis indicates, however, that this branch exhibits exactly one fold bifurcation in the interval $(0, \kappa_1)$, and that the resulting stable branch does not undergo further secondary bifurcations.

These observations complement Theorem 2.3, which ensures existence of a stable nonconstant stationary solution. Numerically, we observe precisely one stable nonconstant solution branch, unique up to spatial shifts due to periodic boundary conditions, suggesting uniqueness of the variational minimizer modulo translations. In the context of the model application, Lyapunov stability is sufficient, as it describes symmetry breaking in systems where the position of the emerging organizer (given by a peak in morphogen concentration) is not prescribed *a priori*. In addition, all unstable branches emerging from bifurcation points $(\bar{\Phi}_n, \kappa_n)$ with $\kappa_n < 1.5$ display an analogous subcritical structure with a single fold bifurcation. Branches associated with different modal numbers remain disjoint for all values of κ .

3. MODEL FORMULATION

We reformulate the mechanochemical patterning model for *Hydra* spheroids introduced in [41] in two spatial dimensions in which the spheroid is described as a closed curve. Starting from the model derivation and formulation in 3D using index notation, one can specialise the notation in 2D as it was done in the supplementary material of [41]. Alternatively, we derive the model directly, using the same modelling arguments but without using index notation, in the Appendix A.

Ultimately, we derive a reduced spatially one-dimensional formulation of the original model in [41] that retains the essential coupling between mechanics and chemistry while simplifying the geometric structure. In this setting, the *Hydra* spheroid is represented as an elastic

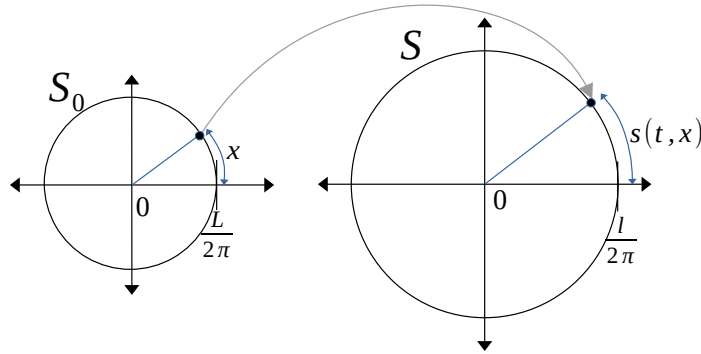


FIGURE 3.1. Schematic representation of the elastic deformation framework for the *Hydra* spheroid model. Left: Circular reference configuration S_0 parametrized by arc-length $x \in [0, L]$. Right: Deformed circular configuration S in which material points have arc-length position $s(x) \in [0, \ell]$.

closed curve, providing a minimal framework to analyze the core mechanochemical feedback mechanisms, see Figure 3.1.

The stress-free reference configuration, denoted by $\mathcal{S}_0$, is assumed to be a circle of fixed circumference $L > 0$. It is parametrised by $x \in [0, L]$ which denotes the arc-length associated to a given point in $\mathcal{S}_0$. We prescribe a fixed enclosed area $A_0 > \text{Area}(\mathcal{S}_0)$ and consider deformations of the reference shape which enclose the area A_0 . In Appendix A we show that, as a consequence of stress balance and the Law of Laplace, the deformed spheroid is always a circle of the given circumference $\ell > L$ and that no other shapes are possible.

The scalar function $s(x) : [0, L] \mapsto [0, \ell]$, $s' > 0$, represents the arc-length position in the deformed spheroid $\mathcal{S}$ of the material point with arc-length position x in $\mathcal{S}_0$. As the deformed spheroid is circular, any deformation is by shifting material points tangentially within the hydra tissue. The full mechanochemical system written in terms of the morphogen concentration $u = u(t, x)$ and the strain $\varepsilon(t, x)$ reads

$$(3.1) \quad \begin{cases} \partial_t u = \kappa \varepsilon - \alpha u + D \partial_{xx} u, \\ 0 = \partial_x (E(u) \varepsilon), \\ \ell - L = \int_0^L \varepsilon dx, \end{cases} \quad \begin{matrix} x \in [0, L], & t \geq 0, \\ & t \geq 0. \end{matrix}$$

This system of equations combines stress balance where $E(u) > 0$ is the elastic modulus given by a monotonically decreasing function of u . This models increased tissue elasticity (reduced stiffness) in response to higher expression of morphogen. This implies that solutions are characterised by larger strain where the morphogen concentration is high.

This "mechanical" equation is coupled to a "chemical" reaction diffusion equation for the morphogen concentration. Here, D denotes diffusivity, α is the spontaneous disassociation rate and $\kappa > 0$ is a given parameter associated to the expression of morphogen in response to tissue strain ε . Up-regulation of morphogen concentration in response to tissue strain complements increased strain where morphogen is over-expressed according to the mechanical part of the model. Together, these two components form a positive local feedback loop.

Note that following infinitesimal stress theory, we write the strain as $\varepsilon = s' - 1$ which motivates the third equation in (3.1) and expresses conservation of total strain, effectively introducing mechanical long-range inhibition, which acts alongside the positive feedback between local strain and morphogen concentration. This combination implements the classical local activation - long-range inhibition (LALI) mechanism.

From the momentum balance equation in (3.1), the stress depends only on time t , that is,

$$\sigma(t) := E(u(t, \xi)) \varepsilon(t, \xi).$$

In particular, the strain can be expressed as

$$\varepsilon(t, \xi) = \frac{\sigma(t)}{E(u(t, \xi))}.$$

Integrating the strain over the whole domain and using the total strain constraint gives

$$\ell - L = \int_0^L \varepsilon(t, \xi) d\xi = \sigma(t) \int_0^L \frac{1}{E(u(t, \xi))} d\xi.$$

Solving for the stress, we obtain

$$\sigma(t) = \frac{\ell - L}{\int_0^L \frac{1}{E(u(t, \xi))} d\xi},$$

and the corresponding strain reads

$$\varepsilon(t, x) = \frac{\sigma(t)}{E(u(t, x))} = \frac{l - L}{E(u(t, x)) \int_0^L \frac{1}{E(u(t, y))} dy}.$$

Using the closed-form expression for the strain, the morphogen equation in (3.1) can be written as

$$(3.2) \quad \partial_t u(t, x) = \frac{\kappa(l - L)}{E(u(t, x)) \int_0^L \frac{1}{E(u(t, y))} dy} - \alpha u + D \partial_{xx} u.$$

For the remainder of the manuscript, we assume that the elastic modulus depends exponentially on the morphogen concentration,

$$E(u) = e^{-\beta u}, \quad \beta > 0.$$

This choice allows (3.2) to be endowed with a variational structure and ensures that the linearization of the spatial operator is self-adjoint.

We next nondimensionalize (3.2) by introducing the scaling

$$t \mapsto t_0 t, \quad \xi \mapsto x_0 x, \quad u \mapsto u_0 u,$$

which leads to

$$\frac{u_0}{t_0} u_t = \kappa \frac{(l - L) e^{\beta u_0 u}}{x_0 \int_0^{L/x_0} e^{\beta u_0 u} dx} - \alpha u_0 u + D \frac{u_0}{x_0^2} u_{xx}.$$

Choosing

$$u_0 = \frac{1}{\beta}, \quad t_0 = \frac{1}{\alpha}, \quad x_0 = L,$$

and defining the dimensionless constants

$$\kappa \frac{\beta(l - L)}{\alpha L} \mapsto \kappa, \quad \frac{D}{\alpha L^2} \mapsto D,$$

we obtain the final nondimensionalized system

$$u_t = D u_{xx} - u + \kappa \frac{e^u}{\int_0^1 e^u dy},$$

for $x \in [0, 1]$, supplemented with periodic boundary conditions

$$u(0, t) = u(1, t), \quad \partial_x u(0, t) = \partial_x u(1, t), \quad t \geq 0.$$

4. STATIONARY SOLUTIONS

4.1. Variational formulation. Equation (2.1) can be interpreted as a gradient flow of the free energy functional with respect to L^2 -norm (see *e.g.* [1]), with

$$(4.1) \quad \mathcal{J}(u(\cdot, t)) = \frac{D}{2} \int_0^1 |u_x(x, t)|^2 dx + \frac{1}{2} \int_0^1 |u(x, t)|^2 dx - \kappa \log \left(\int_0^1 e^{u(x, t)} dx \right).$$

The first and second variations of $\mathcal{J}$ are given by

$$(4.2) \quad \begin{aligned} D\mathcal{J}(u(\cdot, t))\psi &= D \int_0^1 u_x(x, t) \psi_x dx + \int_0^1 u(x, t) \psi dx - \kappa \frac{\int_0^1 e^{u(x, t)} \psi dx}{\int_0^1 e^{u(x, t)} dx}, \\ D^2\mathcal{J}(u(\cdot, t))(\psi, \xi) &= D \int_0^1 \psi_x \xi_x dx + \int_0^1 \psi \xi dx \\ &\quad - \kappa \left(\frac{\int_0^1 e^{u(x, t)} \psi \xi dx}{\int_0^1 e^{u(x, t)} dx} - \frac{\int_0^1 e^{u(x, t)} \psi dx \int_0^1 e^{u(x, t)} \xi dx}{\left(\int_0^1 e^{u(x, t)} dx \right)^2} \right). \end{aligned}$$

The focus of this section is to analyze solutions to equation (2.2) through the critical points of the functional $\mathcal{J} : W^{1,2}(\mathbb{T}) \rightarrow \mathbb{R}$, defined by formula (4.1). A direct analysis of this functional can be challenging due to the difficulty in establishing the existence of nonconstant minimizers. To overcome these difficulties, we decompose the function $\varphi \in W^{1,2}(\mathbb{T})$ using the orthonormal eigenbasis of the Laplace operator with periodic boundary conditions.

Proposition 4.1. *There exist $\{f_k\}_{k=0}^\infty$ an orthonormal basis of $L^2(\mathbb{T})$ and an orthogonal basis of $W^{1,2}(\mathbb{T})$ and a set of eigenvalues $\{\mu_k\}_{k=0}^\infty \subseteq [0, \infty)$, such that (f_k, μ_k) is the eigenpair of the operator Δ*

$$\Delta f_k = -\mu_k f_k,$$

with periodic boundary conditions.

Remark 4.2. For $\mathbb{T} = [0, 1]$ with periodic boundary conditions, the eigenpairs of the Laplacian are given by

$$\begin{aligned} \{f_k\}_{k=0}^\infty &= \{1\} \cup \{\sqrt{2} \sin(2k\pi x) : k \in \mathbb{N}\} \cup \{\sqrt{2} \cos(2k\pi x) : k \in \mathbb{N}_+\}, \\ \{\mu_k\}_{k=0}^\infty &= \{4\pi^2 k^2 : k \in \mathbb{N}\} \end{aligned}$$

These eigenfunctions are uniformly bounded in L^∞ -norm, with $\|f_k\|_\infty = \sqrt{2}$ for $k \geq 1$ and $\|f_0\|_\infty = 1$.

Using the eigenfunction decomposition of the Laplacian, we derive an equivalent system of algebraic equations satisfied by the Fourier coefficients of a stationary solution. Expressing the solution $U(x)$ to equation (2.2) in the eigenbasis as $U = \sum_{k=0}^\infty \bar{R}_k f_k$, with $\bar{R} \in \ell^2(\mathbb{N})$, we obtain

$$(4.3) \quad 0 = -D \sum_{k=0}^\infty \bar{R}_k \mu_k f_k - \sum_{k=0}^\infty \bar{R}_k f_k + \kappa \frac{e^{\bar{R}_0 + \sum_{k=1}^\infty \bar{R}_k f_k}}{\int_0^1 e^{\bar{R}_0 + \sum_{k=1}^\infty \bar{R}_k f_k}}.$$

We multiply equation (4.3) by f_j and integrate over domain $\mathbb{T}$. Then, dividing both the numerator and denominator by $e^{\bar{R}_0}$ and utilizing the orthonormality of the eigenbasis $\{f_j\}_{j=0}^\infty$, we obtain the following system of equations:

$$0 = -\bar{R}_j(D\mu_j + 1) + \kappa \frac{\int_0^1 e^{\sum_{k=1}^\infty \bar{R}_k f_k} f_j}{\int_0^1 e^{\sum_{k=1}^\infty \bar{R}_k f_k}}, \quad \text{for } j \in \mathbb{N}$$

which we rewrite in the form

$$(4.4) \quad \bar{R}_j = \frac{\kappa}{1 + D\mu_j} \frac{\int_0^1 e^{\sum_{k=1}^\infty \bar{R}_k f_k} f_j}{\int_0^1 e^{\sum_{k=1}^\infty \bar{R}_k f_k}}, \quad \text{for } j \in \mathbb{N}.$$

Remark 4.3. The right-hand side of equation (4.4) is bounded and converges to 0 as $j \rightarrow \infty$. Indeed, we have $\|\exp(\sum \bar{R}_k f_k) f_j\|_{L^1} \leq \|f_j\|_{L^\infty} \|\exp(\sum \bar{R}_k f_k)\|_{L^1}$. Since the eigenfunctions are uniformly bounded with $\|f_j\|_\infty = \sqrt{2}$ it follows that

$$|\bar{R}_j| \leq \frac{\sqrt{2}\kappa}{1 + D\mu_j} \xrightarrow{j \rightarrow \infty} 0.$$

To facilitate the analysis of nonconstant solutions to problem (2.2), we introduce a discrete counterpart of the functional $\mathcal{J}$ given by formula (4.1), which offers a more convenient framework for establishing the existence of nonconstant solutions. We express the function $\varphi(x) \in W^{1,2}(\mathbb{T})$ in the eigenbasis, then it holds

$$(4.5) \quad \mathcal{J}(\varphi) = \frac{1}{2}R_0^2 - \kappa R_0 + \sum_{k=1}^{\infty} \frac{R_k^2}{2(1 + D\mu_k)} - \kappa \log \left(\int_0^1 e^{\sum_{k=1}^{\infty} R_k \mu_k} dx \right),$$

where $\varphi(x) = \sum_{k=0}^{\infty} R_k f_k(x)$ with $R \in \omega_\mu^2$ and

$$\omega_\mu^2 = \left\{ R = (R_0, R_1, \dots) \in \ell^2(\mathbb{N}_0) : \|R\|_{\omega_\mu^2} := \sum_{k=0}^{\infty} R_k^2 (1 + D\mu_k) < \infty \right\}.$$

The second derivative of $\mathcal{J}$ given by equation (4.2) and expressed in the eigenbasis, takes the form

$$(4.6) \quad \mathcal{J}_{f_0 f_0}(\varphi) = 1 \quad \text{and} \quad \mathcal{J}_{f_i f_i}(\varphi) = \mathcal{J}_{f_i f_0}(\varphi) = 0 \quad \text{for } i \in \mathbb{N}_+,$$

together with

$$(4.7) \quad \begin{aligned} \mathcal{J}_{f_i f_i}(\varphi) &= 1 + D\mu_i - \kappa \int_0^1 f_i^2(x) p(x) + \kappa \left(\int_0^1 f_i(x) p(x) \right)^2, \\ \mathcal{J}_{f_i f_j}(\varphi) &= -\kappa \int_0^1 f_i(x) f_j(x) p(x) + \kappa \left(\int_0^1 f_i(x) p(x) \right) \left(\int_0^1 f_j(x) p(x) \right), \end{aligned}$$

where $p(x) = e^\varphi / \int_0^1 e^\varphi$ defines a probability measure on $(0, 1)$. Thus by L^∞ -bounds of the eigenfunctions and Jensen inequality we immediately obtain

$$(4.8) \quad 1 + D\mu_i - 2\kappa \leq \mathcal{J}_{f_i f_i}(\phi) \leq 1 + D\mu_i \quad \text{and} \quad |\mathcal{J}_{f_i f_j}(\phi)| \leq 4\kappa$$

for all $i, j \in \mathbb{N}_+$.

We recall two well established properties of the functional $\mathcal{J}$ defined in (4.1). These are included for completeness of the exposition.

Lemma 4.4. *The functional $\mathcal{J} : W^{1,2}(\mathbb{T}) \rightarrow \mathbb{R}$ is bounded from below and coercive.*

Proof. The proof follows directly. By setting $C = \min(D, 1)/2$ and utilizing the embedding $W^{1,2}(\mathbb{T}) \subseteq L^\infty(\mathbb{T})$, we obtain

$$\begin{aligned} J(\varphi) &= \frac{D}{2} \int_0^1 (\varphi_x)^2 + \frac{1}{2} \int_0^1 \varphi^2 - \kappa \log \left(\int_0^1 e^\varphi \right) \\ &\geq C \|\varphi\|_{W^{1,2}}^2 - \kappa \|\varphi\|_{L^\infty} \\ &\geq C \|\varphi\|_{W^{1,2}} (\|\varphi\|_{W^{1,2}} - \kappa/C) \end{aligned}$$

and $J(\varphi) \rightarrow \infty$ as $\|\varphi\|_{W^{1,2}} \rightarrow \infty$. □

Lemma 4.5. *The functional $\mathcal{J} : W^{1,2}(\mathbb{T}) \rightarrow \mathbb{R}$ is weakly lower semicontinuous.*

Proof. Let $\varphi_n \in W^{1,2}(\mathbb{T})$ be a sequence converging $\varphi_n \rightharpoonup \varphi$ weakly in $W^{1,2}$. Clearly

$$\begin{aligned} \int_0^1 |\partial_x \varphi|^2 dy &\leq \liminf_{n \rightarrow \infty} \int_0^1 |\partial_x \varphi_n|^2 dy, \\ \int_0^1 |\varphi|^2 dy &\leq \liminf_{n \rightarrow \infty} \int_0^1 |\varphi_n|^2 dy. \end{aligned}$$

Moreover by compact embedding, the sequence $\varphi_n \rightarrow \varphi$ converges strongly in L^2 and hence by Egorov theorem for each $\varepsilon > 0$ there exist $U \subseteq \mathbb{T}$ such that $|\mathbb{T} \setminus U| < \varepsilon$ and $\varphi_n(x) \rightarrow \varphi(x)$ uniformly on U . By Sobolev embedding the sequence φ_n is uniformly L^∞ -bounded providing

$$\log \left(\int e^{\varphi_n} \right) \rightarrow \log \left(\int e^\varphi \right) \quad \text{as } n \rightarrow \infty$$

and hence $\mathcal{J}(\varphi) \leq \liminf \mathcal{J}(\varphi_n)$. $\square$

4.2. Geometry of the functional. We examine the critical points of the energy functional $\mathcal{J}$ given by the formula (4.1). A constant solution $\bar{U} \equiv \kappa$ is a critical point of $\mathcal{J}$ with $\mathcal{J}(\bar{U}) = -\kappa^2/2$. We demonstrate that, under certain conditions, such a constant solution is not a global minimizer of $\mathcal{J}$.

Lemma 4.6. *For each $\kappa > 0$ there exist a constant $D_{\min}(\kappa)$ such that for all $D < D_{\min}(\kappa)$, the infimum of the functional $\mathcal{J}$ is strictly less than $-\kappa^2/2$.*

Proof. We proceed by considering two distinct cases. For an arbitrary $\kappa > 0$, we first construct a vector $\hat{R}^N \in \mathbb{R}^N$ defined as

$$\hat{R}^N = \left(\kappa, \frac{\sqrt{2}\kappa}{1 + D\mu_1}, \dots, \frac{\sqrt{2}\kappa}{1 + D\mu_N}, 0, 0, \dots \right).$$

This vector is a finite-dimensional approximation of the maximal possible values of $\bar{R}_j$, as noted in Remark 4.3. Let us consider the corresponding function $\varphi_N(x)$ expressed in terms of the eigenbasis $\{\mu_k(x)\}_{k=0}^\infty$, where the cosine components are given by $\hat{R}^N$ and the sine components vanish:

$$\varphi_N(x) = \sum_{k=0}^N \hat{R}_k^N \hat{f}_k(x) \quad \text{with} \quad \hat{f}_k(x) = \sqrt{2} \cos(2\pi kx), \quad \hat{f}_0 = 1.$$

We demonstrate that $\mathcal{J}(\varphi_N) < -\kappa^2/2$ for sufficiently large N . From equation (4.5), we have

$$(4.9) \quad \mathcal{J}(\varphi_N) = -\frac{\kappa^2}{2} + \kappa^2 \sum_{k=1}^N \frac{1}{1 + D\mu_k} - \kappa \log \left(\int_0^1 e^{\sum_{k=1}^N \frac{\sqrt{2}\kappa}{1 + D\mu_k} \hat{f}_k} dx \right).$$

Now we apply a rough estimate for the functions $\hat{f}_k(x) = \sqrt{2} \cos(2\pi kx)$. For each $k \in \{1, \dots, N\}$ we have

$$\hat{f}_k(x) \geq 1 \quad \text{for } x \in [0, 1/(8k)] \cup [1 - 1/(8k), 1].$$

Consequently, $\hat{f}_k(x) \geq 1$ for $x \in [0, 1/(8N)] \cup [1 - 1/(8N), 1]$. This allows us to derive the lower bound

$$\int_0^1 e^{\sum_{k=1}^N \frac{\sqrt{2}\kappa}{1 + D\mu_k} \hat{f}_k} dx \geq 2 \int_0^{1/8N} e^{\sum_{k=1}^N \frac{\sqrt{2}\kappa}{1 + D\mu_k} \hat{f}_k} dx \geq e^{\sum_{k=1}^N \frac{\sqrt{2}\kappa}{1 + D\mu_k}} \frac{1}{4N}$$

which yields

$$(4.10) \quad \kappa \log \left(\int_0^1 e^{\sum_{k=1}^N \frac{\sqrt{2}\kappa}{1 + D\mu_k} \hat{f}_k} dx \right) \geq \kappa^2 \sum_{k=1}^N \frac{\sqrt{2}}{1 + D\mu_k} - \kappa \log(4N).$$

Applying inequality (4.10) to equation (4.9), we obtain

$$\mathcal{J}(\varphi_N) \leq -\frac{1}{2}\kappa^2 - \kappa^2 \sum_{k=1}^N \frac{\sqrt{2}-1}{1+D\mu_k} + \kappa \log(4N) \leq -\frac{1}{2}\kappa^2 - \kappa^2(\sqrt{2}-1) \frac{N}{1+D\mu_N} + \kappa \log(4N).$$

To proceed with our analysis, we examine the conditions ensuring that

$$(4.11) \quad -\frac{1}{2}\kappa^2 - \kappa^2(\sqrt{2}-1) \frac{N}{1+D\mu_N} + \kappa \log(4N) < -\frac{1}{2}\kappa^2$$

holds true for some $N \in \mathbb{N}$. Through algebraic manipulation, inequality (4.11) can be equivalently expressed as

$$(4.12) \quad D < \frac{\kappa(\sqrt{2}-1)N - \log(4N)}{\mu_N \log(4N)}$$

We observe that the right-hand side of inequality (4.12) converges to zero as $N \rightarrow \infty$ and is positive for sufficiently large N , thus ensuring the existence of a global maximum. Accordingly, we define

$$D_1(\kappa) := \max_{N \in \mathbb{N}} \frac{\kappa N(\sqrt{2}-1) - \log(4N)}{\mu_N \log(4N)}.$$

Consequently, for any $D < D_1(\kappa)$, we obtain the strict inequality $\mathcal{J}(\varphi_N) < -\kappa^2/2$.

An alternative approach to establish that the infimum is smaller than $\mathcal{J}(\bar{U})$ involves analyzing the convexity properties of the functional $\mathcal{J}$ at the constant solution. From formula (4.7), the second derivative at $\bar{U} \equiv \kappa$ is expressed as $J_{f_1 f_1}(\bar{U}) = 1 + D\mu_1 - \kappa$. When $\kappa > 1$ and $D < D_2(\kappa) := (\kappa - 1)/\mu_1$, the functional exhibits strict concavity in the direction of f_1 , leading again to $\inf \mathcal{J}(\varphi) < -\kappa^2/2$. We subsequently define

$$D_{\min}(\kappa) = \begin{cases} D_1(\kappa), & \kappa \in (0, 1], \\ \max(D_1(\kappa), D_2(\kappa)), & \kappa > 1, \end{cases}$$

which ensures that $\inf \mathcal{J} < \mathcal{J}(\kappa)$ whenever $D < D_{\min}$. $\square$

Remark 4.7. In Lemma 4.6, we introduced two critical diffusion values, D_1 and D_2 , to characterize distinct behavioral regimes of the functional $\mathcal{J}$. For $D > D_2$ (or for any D when $\kappa \leq 1$), the constant function $\bar{U}(x) \equiv \kappa$ is a local minimum of $\mathcal{J}$. However, in the regime where $D_1 > D_2$ (which occurs for sufficiently small values of κ), the functional may admit multiple local minima, corresponding to coexisting stable steady states (see Remark 5.11).

Next we deal with the case when the diffusion is large.

Lemma 4.8. *For each $\kappa > 0$ there exist a constant $D_{\max}(\kappa)$ such that if $D > D_{\max}(\kappa)$ then the functional $\mathcal{J}$ is globally convex.*

Proof. It is sufficient to prove that the second derivative $D^2 \mathcal{J}(\varphi)(\psi, \psi)$ given by the formula (4.2) is nonnegative for all $\varphi, \psi \in W^{1,2}(\mathbb{T})$ (see e.g. [9, Corollary 3.8.6]). First, fix functions $\varphi, \psi \in W^{1,2}(\mathbb{T})$ and set

$$\psi = \sum_{k=0}^{\infty} R_k f_k(x) \quad \text{and} \quad \psi_n = \sum_{k=0}^n R_k f_k(x) \quad \text{with} \quad R = (R_0, \dots) \in \omega_{\mu}^2.$$

Since $\{f_k\}_{k=0}^{\infty}$ is an orthonormal basis of $L^2(\mathbb{T})$ and an orthogonal basis of $W^{1,2}(\mathbb{T})$, we have $\psi_n \rightarrow \psi$ in $L^2(\mathbb{T})$ and $W^{1,2}(\mathbb{T})$. In particular, by embedding $W^{1,2}(\mathbb{T}) \subseteq L^{\infty}(\mathbb{T})$ there holds

$$\sup_{n \in \mathbb{N}_0} \|\psi_n\|_{L^{\infty}} \leq C \sup_{n \in \mathbb{N}_0} \|\psi_n\|_{W^{1,2}} \leq C \|\psi\|_{W^{1,2}} < \infty$$

which implies

$$\left| \int_0^1 e^\varphi (\psi_n^2 - \psi^2) dx \right| \leq e^{\|\varphi\|_{L^\infty}} (\|\psi_n\|_{L^2}^2 - \|\psi\|_{L^2}^2) \leq C \|\psi_n - \psi\|_{L^2} \|\psi_n + \psi\|_{L^2} \leq C \|\psi_n - \psi\|_{L^2}.$$

Similarly using $\|\psi\|_{L^1} \leq \|\psi\|_{L^2}$ we obtain

$$\left| \left(\int_0^1 e^\varphi \psi_n dx \right)^2 - \left(\int_0^1 e^\varphi \psi dx \right)^2 \right| \leq e^{2\|\varphi\|_{L^\infty}} (\|\psi_n\|_{L^1}^2 - \|\psi\|_{L^1}^2) \leq C \|\psi_n - \psi\|_{L^2}.$$

Now since $\psi_n \rightarrow \psi$ in $W^{1,2}(\mathbb{T})$ passing to the limit yields $D^2\mathcal{J}(\varphi)(\psi_n, \psi_n) \rightarrow D^2\mathcal{J}(\varphi)(\psi, \psi)$ and hence it is enough to prove $D^2\mathcal{J}(\varphi)(\psi_n, \psi_n) \geq 0$ for all $n \in \mathbb{N}_0$.

For each $n \in \mathbb{N}$, the derivative defines a symmetric quadratic form $D^2\mathcal{J}(\varphi)(\psi_n, \psi_n) = \langle H^n R, R \rangle$ for $R \in \mathbb{R}^{n+1}$. The coefficients of the associated matrix $H^n(\varphi) = (\mathcal{J}_{f_i f_j}(\varphi))_{i,j=0}^n \in \mathbb{R}^{(n+1) \times (n+1)}$ are given by formulas (4.6) and (4.7) and by inequalities (4.8) they satisfy $h_{ii} \geq 1 + D\mu_i - 2\kappa$ and $|h_{ij}| \leq 4\kappa$ for all $i, j \in \mathbb{N}$. Since $H^n(\varphi)$ is symmetric, Sylvester's criterion dictates that it suffices to show

$$\det(H^n(\varphi)) > 0 \quad \text{for each } n \in \mathbb{N}_0$$

in order to obtain that $H^n(\varphi)$ is positive definite for all $n \in \mathbb{N}_0$. As an immediate consequence of (4.8) we have $\det(H^0(\varphi)) = 1$ and $\det(H^n(\varphi)) = \det(\mathcal{J}_{f_i f_j}(\varphi))_{i,j=1}^n$. In the following we assume that diffusion is sufficiently large, namely $1 + D\mu_1 - 2\kappa > 0$, implying $h_{ii} > 0$ for all $i \in \mathbb{N}$. Let us define a family of subsets $\mathcal{A}_k = \{A \subset \{1, \dots, n\} : |A| = k\}$. Using the Leibniz formula for determinants and the estimates for the coefficients of $H^n(\varphi)$, we can write

$$(4.13) \quad \det(H^n(\varphi)) = \sum_{\sigma \in S^n} \left(\text{sgn}(\sigma) \prod_{i=1}^n h_{i, \sigma(i)}^n \right) \geq \prod_{i=1}^n h_{ii} - \sum_{k=1}^n (4\kappa)^k k! \left(\sum_{A \in \mathcal{A}_{n-k}} \prod_{j \in A} h_{jj} \right).$$

To obtain the estimate, we count with k the number of off-diagonal elements in $\prod_{i=1}^n h_{i, \sigma(i)}^n$ for some $\sigma \in S^n$ and group permutations with the same number of off-diagonals together. The first term $\prod_{i=1}^n h_{ii}$ corresponds to $\sigma = id$ and thus $k = 0$. The product of k off-diagonal elements is estimated by $(4\kappa)^k$ and multiplied by $k!$ as there are $k!$ possibilities to choose the order of these k off-diagonal elements. The remaining term $\sum_{A \in \mathcal{A}_{n-k}} \prod_{j \in A} h_{jj}$ corresponds to

the sum of all possible choices of building the product of $n - k$ diagonal elements. Dividing equation (4.13) by the product of all diagonal elements we obtain that the determinants are positive if

$$1 > \sum_{k=1}^n (4\kappa)^k k! \left(\sum_{A \in \mathcal{A}_k} \prod_{j \in A} \frac{1}{h_{jj}} \right) \quad \text{for all } n \in \mathbb{N}.$$

Since by inequalities (4.8), the diagonal elements are bounded from below, we obtain that for sufficiently large D , the series $\sum (h_{ii})^{-1}$ is bounded

$$\sum_{i=1}^n \frac{1}{h_{ii}} \leq \sum_{k=1}^{\infty} \frac{1}{(1 + D\mu_k - 2\kappa)} = C(D) \xrightarrow{D \rightarrow \infty} 0.$$

Hence, it holds

$$\sum_{A \in \mathcal{A}_k} \prod_{j \in A} \frac{k!}{h_{jj}} \leq \left(\sum_{i=1}^n \frac{1}{h_{ii}} \right)^k \leq C(D)^k.$$

Choosing $D_{max} > (2\kappa - 1)/\mu_1$ sufficiently large such that $C(D_{max}) < (8\kappa)^{-1}$, we obtain

$$\sum_{k=1}^n (4\kappa)^k \cdot \left(\sum_{A \in \mathcal{A}_k} \prod_{j \in A} \frac{k!}{h_{jj}} \right) \leq \sum_{k=1}^n \left(\frac{4\kappa}{8\kappa} \right)^k < 1 \quad \text{for all } D > D_{max}.$$

Thus, for $D > D_{max}$, the matrix $H^n(\varphi)$ is positive definite for all $n \in \mathbb{N}_0$ and for all $\varphi \in W^{1,2}(\mathbb{T})$. It follows $D^2\mathcal{J}(\varphi)(\psi, \psi) \geq 0$ for all $\varphi, \psi \in W^{1,2}(\mathbb{T})$ for $D > D_{max}$ and the functional $\mathcal{J}$ is globally convex. $\square$

Remark 4.9. Assume $D > a\kappa$ where $a > 0$ is a constant to be determined later on. Then we have

$$\sum_{k=1}^{\infty} \frac{1}{1 + D\mu_k - 2\kappa} \leq \sum_{k=1}^{\infty} \frac{1}{1 + \kappa(a\mu_k - 2)} \leq \frac{1}{\kappa} \sum_{k=1}^{\infty} \frac{1}{(a\mu_k - 2)}.$$

Here we assume $a > 2/\mu_1$ to ensure that all terms are positive and the series converges. By the direct computation we obtain that

$$\sum_{k=1}^{\infty} \frac{1}{(a\mu_k - 2)} = \frac{1}{4} \left(1 - \frac{\sqrt{2\pi} \cot\left(\frac{\sqrt{2\pi}}{\sqrt{a}}\right)}{\sqrt{a}} \right)$$

and the condition $\sum_{k=1}^{\infty} (a\mu_k - 2)^{-1} \leq 1/8$ is satisfied for $a > 15/\mu_1$. This means, we could choose $D_{max}(\kappa) = 15\kappa/\mu_1$ in Lemma 4.8.

4.3. Existence of spatial patterns. We finish this section by proving the existence of nonconstant solutions to problem (2.2) by incorporating the geometry of the functional $\mathcal{J}$. It is clear that each critical point of the functional $\mathcal{J}$ corresponds to a weak solution of problem (2.2).

Proof of Theorem 2.3. First we show the existence of nonconstant stationary solutions. Combining Lemma 4.4 and Lemma 4.5 with a classical extreme value theorem, [11, Thm 6.2.8], we conclude that there exists a minimizer $U \in W^{1,2}(\mathbb{T})$ satisfying $\inf \mathcal{J} = \mathcal{J}(U)$ and the weak formulation of (2.2). By Lemma 4.6 we immediately obtain $\mathcal{J}(U) < \mathcal{J}(\bar{U})$ for $D < D_{min}$ and since there exists only one constant critical point of $\mathcal{J}$ (only one constant solution to equation (2.2)) we immediately conclude that $U(x)$ is nonconstant. As U is a local minimum of $\mathcal{J}$, the steady state U is Lyapunov stable in $L^2(\mathbb{T})$.

Next we show the uniqueness of the stationary solutions for $D > D_{max}$. As by Lemma 4.8 the functional $\mathcal{J}$ is globally convex, every local minimum is already a global minimum. Furthermore, the set of all minima forms a convex set and is, in particular, connected (see e.g. [9, Theorem 3.10.1]). Since $\bar{U} \equiv \kappa$ is a critical point of $\mathcal{J}$, formula (4.2) together with Poincare inequality $\mu_1 \int_0^1 (\psi - \int \psi)^2 \leq \int \psi_x^2$ yields

$$\begin{aligned} D^2\mathcal{J}(\bar{U})(\psi, \psi) &= D \int_0^1 \psi_x^2 + \int_0^1 \psi^2 - \kappa \int_0^1 \psi^2 + \kappa \left(\int_0^1 \psi \right)^2 \\ &\geq (1 + D\mu_1 - \kappa) \int_0^1 \psi^2 + (\kappa - D\mu_1) \left(\int_0^1 \psi \right)^2 \\ &\geq \int_0^1 \psi^2 + (D\mu_1 - \kappa) \left(\int \psi^2 - \left(\int \psi \right)^2 \right). \end{aligned}$$

Hence since $0 \leq \left(\int \psi^2 - (\int \psi)^2 \right) \leq \int \psi^2$ we have

$$D^2 \mathcal{J}(\bar{U})(\psi, \psi) \geq \begin{cases} \int \psi^2 & \text{for } (D\mu_1 - \kappa) > 0, \\ (1 + D\mu_1 - \kappa) \int \psi^2 & \text{for } (D\mu_1 - \kappa) < 0. \end{cases}$$

Since we consider $D > D_{max}$, it is clear that $1 + D\mu_1 - \kappa > 0$. We will later see in Proposition 5.1 that the constant steady state $\bar{U}$ is stable exactly when $1 + D\mu_1 - \kappa > 0$ holds. Thus, this condition is really necessary to prove convexity of $\mathcal{J}$. By the strict positivity of $D^2 \mathcal{J}(\bar{U})(\psi, \psi)$, it follows that $\mathcal{J}$ is strictly convex in a small neighborhood of $\bar{U}$ and in this neighborhood, $\bar{U}$ is the unique minimum of $\mathcal{J}$. Since in case of multiple minima, they would be connected, $\bar{U}$ is the unique critical point of $\mathcal{J}$ and hence a unique solution to equation (2.2). Moreover, as the unique critical point and global minimum of $\mathcal{J}$, the constant steady state $\bar{U}$ is globally asymptotically stable in $L^2(\mathbb{T})$. $\square$

5. LINEAR STABILITY ANALYSIS

In this section, we examine the linear stability of stationary solutions to problem (2.1), which allows us to identify dynamically admissible patterns. Let U be a solution of equation (2.2). To determine the linear stability of U , we insert $u(x, t) = U(x) + e^{\nu t} \varphi(x)$ into (2.1) and linearize about U , which leads to the eigenvalue problem

$$(5.1) \quad D\varphi_{xx} + \left(\kappa \frac{e^U}{\int_0^1 e^U dy} - 1 - \nu \right) \varphi = \kappa \frac{e^U}{(\int_0^1 e^U dy)^2} \int_0^1 e^U \varphi dy$$

supplemented with the boundary conditions $\varphi(0) = \varphi(1)$ and $\varphi_x(0) = \varphi_x(1)$. Throughout our analysis, we employ the notation ν to denote the eigenvalues of this nonlocal problem. Later, we introduce λ to represent the eigenvalues of the corresponding local problem obtained by setting the right-hand side of (5.1) to zero.

5.1. Constant steady state. First we consider stability of the constant steady state. Clearly, $\bar{U} \equiv \kappa$ is the unique constant steady state of problem (2.2) which exists for all parameters D, κ . To investigate its stability, we insert $\bar{U} \equiv \kappa$ in (5.1) and obtain the problem

$$(5.2) \quad D\varphi_{xx} + (\kappa - 1 - \nu) \varphi = \kappa \int_0^1 \varphi dy.$$

Proposition 5.1. *The constant solution $\bar{U} \equiv \kappa$ is stable for $\kappa < 1 + \mu_1 D$ and unstable for $\kappa > 1 + \mu_1 D$.*

Proof. The spectrum of (5.2) is fully characterized by the set of eigenfunctions of the Laplace operator (see Proposition 4.1). Choosing $\varphi \in \{f_k\}_{k=0}^\infty$ and using the fact that $\int_0^1 f_k(x) dx = 0$ for $k \geq 1$, we find that the eigenvalues of the linearized nonlocal operator (5.2) are

$$\nu_k = \begin{cases} -1 & \text{for } k = 0, \\ \kappa - 1 - D\mu_k & \text{for } k \geq 1. \end{cases}$$

Notice, that since $\nu_1 > \nu_k$ for all $k \geq 2$, the claim follows. $\square$

5.2. Local and nonlocal Sturm-Liouville. We now turn our attention to nonconstant steady states U . By Proposition 2.1, we know that $U \in W^{2,2}(\mathbb{T})$. To analyze the spectrum, we compare the eigenvalues of the linearized operator with those arising from the associated local periodic Sturm-Liouville problem. Our approach builds upon the methods developed in [4, 7], which we adapt to the setting of periodic boundary conditions.

In equation (5.1), introducing the notation

$$(5.3) \quad A(x) = \kappa \frac{e^{U(x)}}{\int_0^1 e^{U(y)} dy} - 1, \quad M = \frac{\kappa}{\left(\int_0^1 e^{U(y)} dy\right)^2} > 0, \quad C(x) = e^{U(x)}$$

transforms the equation into the eigenvalue problem

$$(5.4) \quad D\varphi_{xx} + (A(x) - \nu)\varphi = MC(x) \int_0^1 C(y)\varphi(y)dy,$$

supplemented with periodic boundary conditions $\varphi(0) = \varphi(1)$ and $\varphi_x(0) = \varphi_x(1)$. We consider the linearization $\mathcal{L}$ around U as operator on $L^2(\mathbb{T})$, i.e.

$$\mathcal{L} : \mathcal{D}(\mathcal{L}) = W^{2,2}(\mathbb{T}) \subset L^2(\mathbb{T}) \rightarrow L^2(\mathbb{T}), \quad \mathcal{L}\varphi = D\varphi_{xx} + A(x)\varphi - MC(x) \int_0^1 C(y)\varphi(y)dy.$$

Note that the functions A and C are bounded in $x \in \mathbb{T}$ as $U \in W^{2,2}(\mathbb{T}) \subset L^\infty(\Omega)$. The stability of U is then determined by the sign of the real part of the eigenvalues by Lemma 5.2: if $\text{Re}(\nu) < 0$ for all eigenpairs (ν, ϕ) of (5.4), then U is linearly stable; conversely, if there exists an eigenvalue ν with $\text{Re}(\nu) > 0$, then U is linearly unstable.

Lemma 5.2. *The spectrum of $\mathcal{L}$ is discrete, i.e. consists only of isolated eigenvalues of the eigenvalue problem (5.4) with finite multiplicities. The eigenvectors of $\mathcal{L}$ are an orthogonal basis of $L^2(\mathbb{T})$. Moreover, the all eigenvalues are real and countable.*

Proof. The operator $\mathcal{L}$ is self-adjoint with respect to the L^2 -scalar product which can be seen using the specific form of the right-hand-side. Thus, all eigenvalues are real. Moreover, $\mathcal{L}$ has a compact resolvent for $\lambda \in \rho(\mathcal{L})$. This is proven by constructing the resolvent $(\lambda I - \mathcal{L})^{-1}$ using the Lax-Milgram Theorem and proving compactness using the compact embedding $W^{1,2}(\mathbb{T}) \hookrightarrow L^2(\mathbb{T})$. Hence, the spectrum of $\mathcal{L}$ consists entirely of isolated eigenvalues. As $L^2(\mathbb{T})$ is a separable Hilbert space, the spectral theorem for compact self-adjoint operators states that a compact, self-adjoint operator has a basis of eigenvectors. Since $\mathcal{L}$ and the compact operator $(\lambda I - \mathcal{L})^{-1}$ share the same eigenvectors, the eigenvectors of $\mathcal{L}$ are a orthogonal basis of $L^2(\mathbb{T})$. $\square$

Problem (5.4) is closely related to the periodic Sturm–Liouville problem, apart from the presence of the nonlocal term on the right-hand side. Indeed, the eigenvalues ν of (5.4) can be related to solutions of the associated Sturm–Liouville problem

$$(5.5) \quad D\psi_{xx} + (A(x) - \lambda)\psi = 0,$$

together with the boundary conditions $\psi(0) = \psi(1)$ and $\psi_x(0) = \psi_x(1)$. To distinguish the eigenpairs of both problems we use (ν, φ) for solutions of (5.4) and (λ, ψ) for solutions of (5.5).

The local periodic Sturm–Liouville problem was extensively studied in the past (see. e.g. [8, 22]). Here we recall the well-known result for completeness of the exposition.

Proposition 5.3. *The solutions of (5.5) are given by pairs $\{(\lambda_n, \psi_n)\}_{n \in \mathbb{N}_0}$, where the set $\{\psi_n\}_{n \in \mathbb{N}_0}$ is an orthonormal basis of $L^2(0, 1)$. The eigenvalues satisfy*

$$\lambda_0 > \lambda_1 \geq \lambda_2 > \dots > \lambda_{2j+1} \geq \lambda_{2j+2} > \lambda_{2(j+1)+1} \geq \lambda_{2(j+1)+2} > \dots$$

The corresponding eigenfunctions ψ_{2j+1}, ψ_{2j+2} have exactly $2j+2$ zeros in the interval $[0, 1]$. In particular ψ_0 does not vanish on $[0, 1]$ and the eigenvalue λ_0 is simple.

If an eigenfunction ψ_n of the local problem (5.5) satisfies $\int_0^1 C(y)\psi_n(y)dy = 0$, then the pair (λ_n, ψ_n) also solves the nonlocal problem (5.4) and λ_n is a nonlocal eigenvalue. Since the ordering of the eigenvalues λ_n is well understood, we first focus on characterizing the eigenvalues of the nonlocal problem that do not coincide with those of the local problem.

Lemma 5.4. *If a nonlocal eigenvalue ν of (5.4) satisfies $\nu \notin \{\lambda_n\}_{n=0}^\infty$, then ν solves*

$$(5.6) \quad \frac{1}{M} = \sum_{n \in \mathbb{N}_0} \frac{\beta_n^2}{\nu - \lambda_n},$$

where $\beta_n = \int_0^1 C(y)\psi_n(y)dy$. Conversely if $\nu \in \mathbb{R}$ solves (5.6), then ν is an eigenvalue of the nonlocal problem (5.4).

Proof. Let (ν, φ) be an eigenpair of the nonlocal problem (5.4). Assume that ν is not a solution of the local problem (5.5), i.e.

$$J := \int_0^1 C(x)\varphi(x)dx \neq 0.$$

We decompose nonlocal eigenfunction φ in terms of the orthonormal eigenbasis $\{\psi_n\}_{n \in \mathbb{N}_0}$ of the local problem $\varphi(x) = \sum a_n \psi_n$ with $(a_n)_{n \in \mathbb{N}_0} \in \ell^2(\mathbb{N}_0)$. Multiplying equation (5.4) by ψ_m and integrating yields

$$\int_0^1 \sum_{n \in \mathbb{N}_0} a_n \psi_n (D(\psi_m)_{xx} + (A(x) - \nu)\psi_m) dx = JM\beta_m,$$

where $\beta_n = \int_0^1 C(x)\psi_n(x)dx$ and we have applied integration by parts twice. Next, adding and subtracting $\lambda_m \psi_m$ and using orthogonality of the basis ψ_n together with equation (5.5) for (λ_m, ψ_m) , we obtain $a_m(\lambda_m - \nu) = JM\beta_m$, which is equivalent to

$$a_m = J \frac{M\beta_m}{\lambda_m - \nu}.$$

Therefore we can express J in the form

$$J = \int_0^1 C(x)\varphi(x)dx = \sum_{n \in \mathbb{N}_0} a_n \beta_n = JM \sum_{n \in \mathbb{N}_0} \frac{\beta_n^2}{\lambda_n - \nu}.$$

Since $J \neq 0$ by assumption, ν solves (5.6). The series converges for each $\nu \neq \lambda_n$ as $\lambda_n \sim n^2$ [7, 22] and β_n is uniformly bounded. This proves one direction of the claim.

Now, we fix $\nu \in \mathbb{R}$ which solves (5.6). Notice that $\nu \neq \lambda_n$ for all $n \in \mathbb{N}_0$ and there exist $\beta_n \neq 0$. We set $a_n = M\beta_n/(\lambda_n - \nu)$ and define the function $\varphi = \sum_{n \in \mathbb{N}_0} a_n \psi_n$ which is nonzero as $\beta_n \neq 0$ for at least one $n \in \mathbb{N}_0$. Substituting φ into the equation (5.4) and again adding and subtracting $\lambda_n \psi_n$ and utilizing (5.5) we obtain

$$M \sum_{n=0}^\infty \beta_n \psi_n = MC(x) \sum_{n=0}^\infty a_n \int_0^1 C(x)\psi_n.$$

Notice that by assumption (5.6) right hand side satisfies

$$\sum_{n=0}^\infty a_n \int_0^1 C(x)\psi_n = \sum_{n=0}^\infty a_n \beta_n = \sum_{n=0}^\infty M \frac{\beta_n^2}{\nu - \lambda_n} = 1$$

and since $\sum \beta_n \psi_n = C(x)$ the proof is completed. $\square$

5.3. Relation between eigenvalues. Next, we establish the location of eigenvalues ν of the nonlocal problem (5.6). First we decompose the set of local eigenvalues into two subsets

$$\{\lambda_n\}_{n=0}^\infty = \{\lambda_n : \beta_n = 0\} \cup \{\lambda_n : \beta_n \neq 0\} := \Lambda_0 \cup \Lambda_\beta$$

We must account for the terms where $\beta_n = 0$ as they vanish in the series (5.6). These sets are not necessarily disjoint, i.e. there can exist a double eigenvalue $\lambda_n = \lambda_{n+1}$ and $\lambda_n \in \Lambda_0$

while $\lambda_{n+1} \in \Lambda_\beta$. Without loss of generality we assume that the elements of Λ_0, Λ_β are in monotonically decreasing order

$$\Lambda_0 = \{\lambda_{n_1}, \lambda_{n_2}, \dots\}, \quad \Lambda_\beta = \{\lambda_{m_1}, \lambda_{m_2}, \dots\}$$

Notice that Λ_0, Λ_β can be infinite, finite or empty and the eigenvalues can be double. Using the introduced notation, equation (5.6) transforms to

$$(5.7) \quad \frac{1}{M} = \sum_{\lambda_{m_k} \in \Lambda_\beta} \frac{\beta_{m_k}^2}{\lambda_{m_k} - \nu}.$$

With this preparation, we can characterize all eigenvalues of the nonlocal problem (5.4).

Theorem 5.5. *The eigenvalues of the nonlocal problem (5.4) are given by*

$$\{\nu_n\}_{n=1}^\infty = \Lambda_0 \cup \{\nu \in \mathbb{R} : \nu \text{ solves equation (5.7)}\}.$$

For all distinct and subsequent local eigenvalues $\lambda_{m_i}, \lambda_{m_{i+1}} \in \Lambda_\beta$ there exists exactly one solution ν of equation (5.7) with $\nu \in (\lambda_{m_{i+1}}, \lambda_{m_i})$. In particular if Λ_β is finite, there exists exactly one solution of equation (5.7) with $\nu \in (-\infty, \min \Lambda_\beta)$.

Proof. It is clear $\Lambda_0 \subseteq \{\nu_n\}_{n=1}^\infty$ and thus the first part follows from Lemma 5.4.

For the second part first note that all solutions ν of (5.7) are real. They can be estimated as follows. Fix subsequent local eigenvalues $\lambda_{m_i}, \lambda_{m_{i+1}} \in \Lambda_\beta$. From now on we assume that λ_{m_i} and $\lambda_{m_{i+1}}$ are distinct. If they coincide, we merge the two summands in the series (5.7) by setting $\beta_{m_i} = \sqrt{\beta_{m_i}^2 + \beta_{m_{i+1}}^2}$.

Then, the right-hand-side of (5.7) is continuous and monotone increasing in $\nu \in (\lambda_{m_{i+1}}, \lambda_{m_i})$ and converging to $-\infty$ as $\nu \downarrow \lambda_{m_{i+1}}$ and to ∞ as $\nu \uparrow \lambda_{m_i}$. Since $M > 0$ by (5.3), there exists a unique solution ν_{m_i} of (5.7) in the interval $(\lambda_{m_{i+1}}, \lambda_{m_i})$. Note that the right-hand-side of (5.7) is negative for all $\nu > \lambda_{m_0}$ and, thus, there exists no solution $\nu > \lambda_{m_0}$. In case of finite Λ_β , there exists a solution ν in the interval $(-\infty, \min \Lambda_\beta)$ as the right-hand-side of (5.7) converges to 0 as $\nu \downarrow -\infty$ and to ∞ as $\nu \uparrow \min \Lambda_\beta$. $\square$

Remark 5.6. Applying this characterization to the eigenvalue problem for the constant steady solution $\bar{U} \equiv \kappa$, we see that $\Lambda_\beta = \{\kappa - 1\}$ and $\Lambda_0 = \{\lambda_k\}_{k=1}^\infty$ and hence the nonlocal eigenvalues satisfy $\nu_0 \in (-\infty, \kappa - 1)$ and $\nu_j = \lambda_j$ for all $j \geq 1$. Indeed, this coincides with the results from Proposition 5.1.

5.4. Stability of steady states. With this characterization, we can find a criterion under which the steady state solution U is (linearly) unstable.

Theorem 5.7. *If the derivative U_x of a stationary solution U has at least three zeros in $[0, 1)$, then U is linearly unstable.*

Proof. Since U solves (2.2), we can differentiate (2.2) with respect to x and obtain that U_x solves

$$0 = D(U_x)_{xx} - U_x + \kappa \frac{e^U}{\int_0^1 e^U dy} U_x = D(U_x)_{xx} + A(x)U_x,$$

which is exactly the local problem (5.5) with $\lambda = 0$. Since $U \in W^{2,2}(\mathbb{T})$, all U , U_x and U_{xx} satisfy periodic boundary conditions. Hence, the local problem has a zero eigenvalue with corresponding eigenfunction with at least 3 zeros on the interval $[0, 1)$. Proposition 5.3 implies that $\lambda_3 \geq 0$ as the eigenfunctions ψ_0, ψ_1, ψ_2 have at most two zeros on the interval $[0, 1)$. Therefore there are at least three strictly positive local eigenvalues $\lambda_0 > \lambda_1 \geq \lambda_2 > \lambda_3 \geq 0$. This is already sufficient to obtain a positive nonlocal eigenvalue ν . Indeed, if all three local eigenvalues are in Λ_β then by Theorem 5.5 there exists a positive nonlocal eigenvalue

$\nu \in (\lambda_0, \lambda_1)$. Otherwise at least one of those local eigenvalues is also a nonlocal eigenvalue (belongs to Λ_0) and hence the proof is finished. $\square$

Remark 5.8. If U is a steady state solution, with U_x having two zeros in the interval $[0, 1]$, then either $\lambda_1 = 0$ or $\lambda_2 = 0$. Even though $\lambda_1 = 0$ implies $\lambda_0 > 0$, this is not enough to obtain instability of the nonlocal problem. Indeed, assuming $\lambda_0 \in \Lambda_\beta$ and $\lambda_1 \in \Lambda_0$ implies $\Lambda_0 \subseteq (-\infty, 0]$. Moreover, by Theorem 5.5 the nonlocal eigenvalues satisfy $\nu_0 = 0$ and $\nu_1 \in (\lambda_{m_1}, \lambda_0)$ with $\lambda_{m_1} \in \Lambda_\beta$. If $\lambda_{m_1} < 0$ it is possible that $\nu_1 < 0$ and hence all nonlocal eigenvalues are nonpositive.

To relate these results to the nonlinear dynamics, we exploit the fact that problem (2.1) can be viewed as a gradient flow in $L^2(\mathbb{T})$ with respect to the Lyapunov functional $\mathcal{J}$ defined in (4.1). This allows us to connect linear instability with nonlinear instability in the L^2 -norm for perturbations in $W^{1,2}(\mathbb{T})$.

Lemma 5.9. *Let U be a linear unstable steady state of (2.1). Then, the steady state is nonlinearly unstable with respect to the L^2 -norm.*

Proof. The problem (2.1) is a gradient flow with respect to the L^2 -norm, thus it follows from the general theory of gradient flows and dynamical systems [1] that nonlinearly stable solutions to problem (2.1) are local minima of the Lyapunov functional $\mathcal{J}$ given by formula (4.1). For the linearization $\mathcal{L}$ at U we have

$$D^2\mathcal{J}(U)(\psi, \psi) = (-\mathcal{L}\psi, \psi)_{L^2} \quad \text{for all } \psi \in W^{2,2}(\mathbb{T}),$$

compare to equation (4.2). Hence, the linearization corresponds to minus Hessian of $\mathcal{J}$ at U . By Lemma 5.2, the linear instability of U implies the existence of a positive real eigenvalue λ . So if $\mathcal{L}$ has an eigenvalue $\lambda > 0$ with eigenvector φ , then

$$D^2\mathcal{J}(U)(\varphi, \varphi) = -\lambda\|\varphi\|_{L^2}^2 < 0.$$

Hence, U is not a local minimum of $\mathcal{J}$ and thus nonlinearly unstable. $\square$

Remark 5.10. In general, linear stability of a steady state does not necessarily imply nonlinear stability in the L^2 -norm; it often only guarantees nonlinear stability in the L^∞ -norm [21], or may require restricting the class of admissible perturbations [3] as we also did in the above lemma.

With these results we can now prove our main result about the instability of m -modal solutions in Theorem 2.4.

Proof of Theorem 2.4. Let U be an arbitrary stationary m -periodic solution to problem (2.1). Then, U has at least m peaks and hence, the derivative U_x has at least $2m - 1$ zeros in the interval $[0, 1]$. By Theorem 5.7, all m -modal solution U are linearly unstable for $m \geq 2$. By Lemma 5.9, nonlinear instability of all m -modal solutions follows.

The periodic solution to problem (2.2) of Theorem 2.3 constructed in Section 4 is a minimum of the Lyapunov functional given by formula (4.1). Therefore, the solution is nonlinearly stable and the solution is unimodal. $\square$

Corollary 5.11 (Bistability). *For $\kappa < 1$ and sufficiently small $D > 0$, the system exhibits bistability: both the constant state $U \equiv \kappa$ (stable by Proposition 5.1) and the nonconstant unimodal solution (stable by Theorems 2.3 and 2.4) coexist.*

6. BIFURCATION ANALYSIS

In this section we study nonconstant steady states to problem (2.1), bifurcating from the branch of constant solutions $\Phi(\kappa) \equiv \kappa$. Here, κ is also a bifurcation parameter. The linearized

operator $\mathcal{L}(\kappa)$ at a constant steady state $\bar{\Phi}(\kappa)$ is given by

$$\mathcal{L}(\kappa)(\varphi) = D\varphi_{xx} + (\kappa - 1)\varphi - \kappa \int_0^1 \varphi(y)dy,$$

and we emphasize its dependency on the bifurcation parameter κ . Spectral properties of the linearized operator were discussed in Section 5.1. We recall, the eigenvalues of $\mathcal{L}(\kappa)$ are

$$\nu_0 = -1, \quad \nu_k = \kappa - 1 - D\mu_k,$$

where $\mu_k = 4\pi^2 k^2$ are the eigenvalues of $-\Delta$ with periodic boundary conditions. The eigenvalue ν_n crosses zero for each critical value $\kappa_n := 1 + D\mu_n$. Thus, the linearized operator $\mathcal{L}(\kappa)$ has a zero eigenvalue for all $\kappa = \kappa_n$. We denote the corresponding constant solution by $\bar{\Phi}_n \equiv \bar{\Phi}(\kappa_n)$.

6.1. Existence of branches. To establish the existence of nonconstant solution branches bifurcating from $(\bar{\Phi}_n, \kappa_n)$, we apply the Crandall–Rabinowitz theorem for bifurcation from simple eigenvalues [18, Theorem I.5.1]. We first define the relevant function spaces:

$$Y := \{\varphi \in L^2(\mathbb{T}) \mid \varphi(x) = \varphi(1-x) \text{ for a. e. } x \in \mathbb{T} : \text{symmetry constraint}\},$$

$$X := \{\varphi \in W^{2,2}(\mathbb{T}) \mid \varphi(x) = \varphi(1-x) \text{ for a. e. } x \in \mathbb{T}\}.$$

Note that the periodic boundary conditions of the Laplace operator Δ are naturally incorporated in the space $W^{2,2}(\mathbb{T})$. The symmetry condition ensures, the kernel of $\mathcal{L}(\kappa_n)$ is one-dimensional:

$$\ker \mathcal{L}(\kappa_n) = \text{span}\{\varphi_n\}, \quad \text{where } \varphi_n := \sqrt{2} \cos(2n\pi x).$$

We consider solutions in a sufficiently small neighborhood of $(\bar{\Phi}_n, \kappa_n)$ of the form

$$\varphi = \bar{\Phi}(\kappa) + \chi, \quad \kappa = \kappa_n + \alpha, \quad \text{with } \chi \in X, \alpha \in \mathbb{R}.$$

Define the mapping $G : X \times \mathbb{R} \rightarrow Y$ by

$$G(\chi, \alpha) := D\chi_{xx} - \chi + (\kappa_n + \alpha) \left(\frac{e^\chi}{\int e^\chi dy} - 1 \right).$$

Then $G(\chi, \alpha) = 0$ if and only if $\varphi = \bar{\Phi}(\kappa_n + \alpha) + \chi$ solves the stationary problem (2.2) for $\kappa = \kappa_n + \alpha$. Note that, the trivial branch of constant steady states $(\bar{\Phi}(\kappa), \kappa)$ transforms to $(0, \alpha)$ and has a zero eigenvalue at $\alpha = 0$.

The map G is well-defined as $X \subset L^\infty(\mathbb{T})$ and thus $G(X \times \mathbb{R}) \subset L^2(\mathbb{T})$. To confirm the symmetry constraint, use that $\chi_x(x) = -\chi_x(1-x)$ if $\chi(x) = \chi(1-x)$ and χ regular enough. Hence, we have $\chi_{xx}(x) = \chi_{xx}(1-x)$ and $G(\chi, \alpha)(x) = G(\chi, \alpha)(1-x)$ follows.

We are now in a position to apply the Crandall–Rabinowitz theorem to the equation $G(\chi, \alpha) = 0$ at the bifurcation point $(0, 0)$.

Theorem 6.1. *For each $n \in \mathbb{N}$ there exist $\delta > 0$ and smooth functions*

$$\alpha_n(-\delta, \delta) \rightarrow \mathbb{R}, \quad z_n : (-\delta, \delta) \rightarrow X$$

with $\alpha_n(0) = z_n(0) = 0$, such that curve

$$\Gamma_n(s) = \left(\bar{\Phi}_n + \alpha_n(s) + s\varphi_n + sz_n(s), \kappa_n + \alpha_n(s) \right)$$

forms a branch of solutions to problem (2.2) bifurcating from $(\bar{\Phi}_n, \kappa_n)$. For all $s \in (-\delta, \delta) \setminus \{0\}$, the solutions on $\Gamma_n(s)$ are nonconstant.

Proof. Let us verify the assumptions of [18, Theorem I.5.1]. The trivial branch condition holds as $G(0, \alpha) = 0$ for all $\alpha \in \mathbb{R}$.

Next, we prove that $G_\chi(0, 0)$ is a Fredholm operator of index zero with one-dimensional kernel. The linearization $G_\chi(0, 0) : X \rightarrow Y$ is given by

$$(6.1) \quad G_\chi(0, 0)(h) = Dh'' + (\kappa_n - 1)h - \kappa_n \int_0^1 h(y) dy.$$

The kernel is one-dimensional as $\ker G_\chi(0, 0) = \ker \mathcal{L}(\kappa_n) = \text{span}\{\varphi_n\}$. To establish that $\text{codim ran } G_\chi(0, 0) = 1$, we prove that $\text{ran } G_\chi(0, 0) = \text{span}\{\varphi_n\}^\perp$.

First, we show $\text{ran } G_\chi(0, 0) \subseteq \text{span}\{\varphi_n\}^\perp$. For any $h \in X$, multiplying (6.1) by φ_n and integrating:

$$\int_0^1 G_\chi(0, 0)(h) \cdot \varphi_n dy = \int_0^1 Dh\varphi_n'' dy + (\kappa_n - 1) \int_0^1 h\varphi_n dy = 0,$$

where we used integration by parts, $\varphi_n'' = -\mu_n\varphi_n$, $\kappa_n - 1 = D\mu_n$, and $\int_0^1 \varphi_n dy = 0$.

For the reverse inclusion, let $g \in Y$ with $\langle g, \varphi_n \rangle = 0$. We seek $h \in X$ such that $G_\chi(0, 0)(h) = g$. Expanding in the eigenbasis of $\mathcal{L}(\kappa_n)$: $g = \sum_{j=0}^\infty g_j \varphi_j$ with $g_n = 0$, we obtain

$$h = -g_0 + \sum_{j \neq n} \frac{g_j}{\kappa_n - 1 - D\mu_j} \varphi_j.$$

This is well-defined since $\kappa_n - 1 - D\mu_j = D(\mu_n - \mu_j) \neq 0$ for $j \neq n$. Furthermore, we only use the terms of the eigenbasis which satisfy the symmetry condition $\varphi(x) = \varphi(1-x)$ as $g \in Y$ and thus by regularization of $(\partial_{xx} - a)^{-1}$ we have $h \in X$. Therefore, $\text{ran } G_\chi(0, 0) = \text{span}\{\varphi_n\}^\perp$ has codimension one. Since the kernel is one-dimensional and the range has codimension one, $G_\chi(0, 0)$ is a Fredholm operator of index zero.

For the transversality condition of the Crandall-Rabinowitz theorem, observe that

$$G_{\chi\alpha}(0, 0)(\varphi_n) = \varphi_n - \int_0^1 \varphi_n dy = \varphi_n \notin \text{ran } G_\chi(0, 0),$$

since $\int_0^1 \varphi_n dy = 0$ for $n \geq 1$.

Having verified all conditions (trivial branch, Fredholm property with index zero, one-dimensional kernel, and transversality condition), we obtain the existence of a branch $(\chi_n(s), \alpha_n(s))$ of solutions satisfying $G(\chi_n(s), \alpha_n(s)) = 0$ for all $s \in (-\delta, \delta)$ with $(\chi_n(0), \alpha_n(0)) = (0, 0)$. More precisely, there exists an open neighborhood $B_1 \subset B$ of $(0, 0)$ and smooth functions $\alpha_n : (-\delta, \delta) \rightarrow \mathbb{R}$, $z_n : (-\delta, \delta) \rightarrow X$ with $\alpha_n(0) = z_n(0) = 0$ such that

$$B_1 \cap G^{-1}(0) = \{(s\varphi_n + sz_n(s), \alpha_n(s)) \mid -\delta < s < \delta\} \cup \{(0, s) \mid -\varepsilon < s < \varepsilon\}.$$

Consequently, $(\bar{\Phi}_n + \alpha_n(s) + s\varphi_n + sz_n(s), \kappa_n + \alpha_n(s))$ constitutes a branch of solutions to problem (2.2). Since φ_n is nonconstant and $z_n(s) \rightarrow 0$ as $s \rightarrow 0$, these solutions are nonconstant for all $s \neq 0$ with sufficiently small $|s|$. $\square$

6.2. Direction of branches (sub- and supercritical pitchfork). To determine the direction of bifurcation, we analyze the behavior of the bifurcation parameter $\alpha_n(s)$ near $s = 0$. Define the function

$$\mathcal{G}(s) := G(s\varphi_n + sz_n(s), \alpha_n(s)) \equiv 0$$

and differentiate it with respect to s to extract the behavior of $\alpha_n(s)$ near the bifurcation point.

Lemma 6.2. *The branch of solutions $\Gamma_n(s) = (\bar{\Phi}_n + \alpha_n(s) + s\varphi_n + sz_n(s), \kappa_n + \alpha(s))$ constructed in Theorem 6.1 has the asymptotic expansion*

$$\Gamma_n(s) \approx \left(\kappa_n + \varphi_n s + (\alpha_n''(0) + z_n'(0))s^2, \kappa_n + \alpha_n''(0)s^2 \right),$$

with

$$z_n'(0) = \frac{\kappa_n}{24\pi^2 n^2 D} \cos(4\pi n x) \quad \text{and} \quad \alpha_n''(0) = \frac{1}{4} - \frac{1}{16Dn^2\pi^2} + 2Dn^2\pi^2.$$

Proof. We differentiate $\mathcal{G}(s)$ with respect to s up to third order. The derivatives of G with respect to χ and α are given in Appendix C.

First derivative: Differentiating $\mathcal{G}(s)$ with respect to s , we obtain

$$G_\chi(s\varphi_n + sz_n(s), \alpha_n(s))(\varphi_n + z_n(s) + sz_n'(s)) + G_\alpha(s\varphi_n + sz_n(s), \alpha_n(s))\alpha_n'(s) = 0.$$

Evaluating at $s = 0$ and using $z_n(0) = 0$, $\alpha_n(0) = 0$, and $G_\alpha(0, 0) = 0$, we obtain

$$G_\chi(0, 0)(\varphi_n) = 0,$$

which holds since $\varphi_n \in \ker G_\chi(0, 0)$. This does not determine $\alpha_n'(0)$, so we proceed to the second derivative.

Second derivative: Differentiating twice and evaluating at $s = 0$, using $z_n(0) = 0$, $\alpha_n(0) = 0$, $\int \varphi_n dy = 0$, $G_\alpha(0, 0) = 0$, and $G_{\alpha\alpha}(\chi, \alpha) \equiv 0$, we obtain

$$\begin{aligned} 0 &= G_{\chi\chi}(0, 0)(\varphi_n, \varphi_n) + 2G_{\chi\alpha}(0, 0)(\varphi_n)\alpha_n'(0) + 2G_\chi(0, 0)(z_n'(0)) \\ &= 2(Dz_n'(0)_{xx} + (\kappa_n - 1)z_n'(0)) - 2\kappa_n \int z_n'(0) dy + \kappa_n \left(\varphi_n^2 - \int \varphi_n^2 dy \right) + 2\varphi_n \alpha_n'(0). \end{aligned}$$

Since $z_n'(0)$ is periodic, we obtain $\int z_n'(0) dy = 0$ and hence, the corresponding equation can be rewritten as

$$(6.2) \quad 2(Dz_n'(0)_{xx} + (\kappa_n - 1)z_n'(0)) = -\kappa_n \left(\varphi_n^2 - \int \varphi_n^2 dy \right) - 2\varphi_n \alpha_n'(0).$$

Multiplying equation (6.2) by φ_n , integrating over the domain and using the identities $\varphi_n'' = -\mu_n \varphi_n$ and $\kappa_n - 1 = D\mu_n$ together with $\int \varphi_n^3 = \int \varphi_n = 0$ we conclude

$$0 = -2\alpha_n'(0) \int \varphi_n^2 dy$$

which implies $\alpha_n'(0) = 0$. To find an explicit expression for $z_n'(0)$, we solve

$$Dz_n'(0)_{xx} + (\kappa_n - 1)z_n'(0) = -\frac{\kappa_n}{2} \left(\varphi_n^2 - \int \varphi_n^2 dy \right) = -\frac{\kappa_n}{2} \cos(4n\pi x),$$

where we used $\varphi_n(x)^2 = 2\cos^2(2\pi n x) = 1 + \cos(4\pi n x)$ and $\int \varphi_n^2 dy = 1$. The solution has the explicit form

$$z_n'(0) = A \cos(4n\pi x), \quad \text{with} \quad -DA(4n\pi)^2 + (\kappa_n - 1)A = -\frac{\kappa_n}{2}.$$

Solving for A and using $\kappa_n - 1 = D\mu_n = D4\pi^2 n^2$ gives $A = \kappa_n(24n^2\pi^2 D)^{-1}$.

Third derivative: Differentiating $\mathcal{G}(s)$ three times and evaluating at $s = 0$, noting that $\int z_n''(0) dy = 0$, in analogy with $\int z_n'(0) dy = 0$ and using the properties derived from the second derivative, we obtain

$$\begin{aligned} (6.3) \quad 0 &= 3(Dz_n''(0)_{xx} + (\kappa_n - 1)z_n''(0)) + \kappa_n \left(\varphi_n^3 - 3\varphi_n \int \varphi_n^2 dy \right) \\ &\quad + 6\kappa_n \left(\varphi_n z_n'(0) - \int \varphi_n z_n'(0) dy \right) + 2\alpha_n''(0)\varphi_n. \end{aligned}$$

Analogously, we multiply equation (6.3) by φ_n , integrate, and use identities

$$\int_0^1 \varphi_n^2 dy = 1 \quad \int_0^1 \varphi_n^4 dy = 3/2 \quad \int_0^1 \varphi_n^2 z'_n(0) dy = 1/2A$$

which by direct calculation yields

$$0 = 3/2\kappa_n - 3\kappa_n + 3\kappa_n A + 2\alpha_n''(0).$$

Substituting $A = \kappa_n(24\pi^2 n^2 D)^{-1}$ and $\kappa_n = 1 + D4\pi^2 n^2$, we obtain

$$(6.4) \quad \alpha_n''(0) = \frac{1}{4} - \frac{1}{16Dn^2\pi^2} + 2Dn^2\pi^2.$$

□

Having derived an explicit formula for $\alpha_n''(0)$, we can now classify the type of bifurcation. We emphasize that this classification applies only to the local behavior near the bifurcation point and is determined solely by the eigenvalue that crosses zero. Other eigenvalues in the spectrum, although not involved in the bifurcation mechanism, do not affect this local characterization. Importantly, the bifurcation type by itself does not determine the overall linear stability of the bifurcating branch; a full stability assessment requires examining the entire spectrum of the linearized operator along the bifurcating solutions.

Theorem 6.3. *Fix $n \in \mathbb{N}$. If $\kappa_n \neq 1.5$, then $(\bar{\Phi}_n, \kappa_n)$ undergoes a pitchfork bifurcation. The type of bifurcation is determined by the sign of $\alpha_n''(0)$:*

- *subcritical if $\alpha_n''(0) < 0$, i.e., $\kappa_n < 1.5$ (equivalently, $0 < D < (8n^2\pi^2)^{-1}$),*
- *supercritical if $\alpha_n''(0) > 0$, i.e., $\kappa_n > 1.5$ (equivalently, $D > (8n^2\pi^2)^{-1}$).*

Proof. Observe that by formula (6.4) there holds $\alpha_n''(0) = 0$ if and only if $D = \frac{1}{8n^2\pi^2}$, which corresponds to $\kappa_n = 1.5$. Thus, whenever $\alpha_n'(0) = 0$ and $\alpha_n''(0) \neq 0$, a pitchfork bifurcation occurs at $\bar{\Phi}_n$ for $\kappa = \kappa_n$.

To determine the type of bifurcation, we examine the behavior of the trivial branch. Since the critical eigenvalue responsible for bifurcation is $\kappa - \kappa_n$, the trivial branch loses stability as κ crosses κ_n . By the principle of exchange of stability, the bifurcation is subcritical if the nontrivial branch enters the region where the trivial solution is stable, i.e., if $\kappa_n + \alpha_n(s) < \kappa_n$ for small $s > 0$. Conversely, it is supercritical if $\kappa_n + \alpha_n(s) > \kappa_n$. Using the expansion $\alpha_n(s) = \frac{1}{2}\alpha_n''(0)s^2 + \mathcal{O}(s^3)$, it follows that $\alpha_n''(0) < 0$ implies a subcritical bifurcation, while $\alpha_n''(0) > 0$ implies a supercritical one. The equivalence between the sign of $\alpha_n''(0)$ and the conditions on D and κ_n follows from direct computation. □

To conclude about the stability of the branch, we consider the bifurcation point κ_1 , namely a point at which the trivial branch changes stability.

Theorem 6.4. *Fix $(\bar{\Phi}_n, \kappa_n)$ for some $n \in \mathbb{N}$, and let $\Gamma_n(s) = (\Phi_n(s), \kappa_n(s))$ denote the bifurcating branch. Then, for sufficiently small $|s|$, the following stability properties hold:*

- *For $n \geq 2$, the branch $\Gamma_n(s)$ consists of linearly unstable solutions;*
- *For $n = 1$ and $\kappa_1 < 1.5$, the branch $\Gamma_1(s)$ consists of linearly unstable solutions;*
- *For $n = 1$ and $\kappa_1 > 1.5$, the branch $\Gamma_1(s)$ consists of linearly stable solutions.*

Proof. At the bifurcation point $(\bar{\Phi}_n, \kappa_n)$ with $n \geq 2$, Proposition 5.1 shows that the linearized operator $\mathcal{L}(\kappa_n)$ possesses exactly $n-1$ positive eigenvalues in addition to the simple zero eigenvalue. By the continuation principle, these positive eigenvalues persist along the bifurcating branch $\Gamma_n(s)$ for sufficiently small $|s|$. Consequently, regardless of the sign of the critical eigenvalue along the branch, the solutions $\Phi_n(s)$ remain linearly unstable in a neighborhood of the bifurcation point.

For the primary bifurcation at $(\bar{\Phi}_1, \kappa_1)$, the linearized operator $\mathcal{L}(\kappa_1)$ has a simple zero eigenvalue and all other eigenvalues are negative. The stability of the bifurcating branch $\Gamma_1(s)$ is therefore determined exclusively by the behavior of the critical eigenvalue along the nontrivial branch. The assertion follows from an analogous argument to that employed in Theorem 6.3. $\square$

6.3. Global continuation and turning points. Fix a bifurcation point $(\bar{\Phi}_n, \kappa_n)$ for some $n \in \mathbb{N}$. We now establish that the local branch $\Gamma_n(s)$ obtained in Theorem 6.1 extends globally.

To apply the global bifurcation theory of Rabinowitz, we reformulate the steady-state equation (2.2) as a fixed-point problem. Specifically, (2.2) is equivalent to

$$0 = \varphi + (D\partial_{xx} - I)^{-1} \left(\kappa \frac{e^\varphi}{\int_{\mathbb{T}} e^\varphi dy} \right) =: \varphi + \kappa f(\varphi) =: H(\varphi, \kappa).$$

Let $\mathcal{S}$ denote the closure of nontrivial solutions to $H(\varphi, \kappa) = 0$ in $W^{1,2}(\mathbb{T}) \times \mathbb{R}$. Note that $\Gamma_n(s) \subset \mathcal{S}$, and by elliptic regularity, $\mathcal{S} \subset W^{2,2}(\mathbb{T}) \times \mathbb{R}$.

Theorem 6.5 (Global bifurcation). *Let $\mathcal{C}$ be the maximal connected component in $\mathcal{S}$ containing $(\bar{\Phi}_n, \kappa_n)$. Then $\mathcal{C}$ satisfies one of the following alternatives:*

- (i) $\mathcal{C}$ is unbounded in $W^{1,2}(\mathbb{T}) \times \mathbb{R}$;
- (ii) $\mathcal{C}$ contains a point $(\bar{\Phi}(\kappa), \kappa)$ on the trivial branch for some $\kappa \neq \kappa_n$.

Proof. We verify the assumptions of the Global Crandall-Rabinowitz Theorem [18, special case of Theorem II.3.3]. Since $W^{1,2}(\mathbb{T}) \subseteq L^\infty(\mathbb{T})$, we have $e^\varphi \in L^\infty(\mathbb{T})$ for all $\varphi \in W^{1,2}(\mathbb{T})$. By elliptic regularity and the compact embedding $W^{2,2}(\mathbb{T}) \subseteq W^{1,2}(\mathbb{T})$, the operator

$$(D\partial_{xx} - I)^{-1} : L^2(\mathbb{T}) \rightarrow W^{2,2}(\mathbb{T}) \subseteq W^{1,2}(\mathbb{T})$$

is compact. Therefore, $f : W^{1,2}(\mathbb{T}) \rightarrow W^{1,2}(\mathbb{T})$ is well-defined and completely continuous.

To establish continuity of the linearization, we compute $D_\varphi H(\bar{\Phi}_n, \kappa) = I + \kappa D_\varphi f(\bar{\Phi}_n)$. At the constant state $\bar{\Phi}_n \equiv \kappa_n$, the linearization yields

$$D_\varphi f(\bar{\Phi}_n)\psi = (D\partial_{xx} - I)^{-1} \left(\psi - \int_{\mathbb{T}} \psi dy \right).$$

Standard elliptic estimates and Sobolev embedding give

$$\|D_\varphi f(\bar{\Phi}_n)\psi\|_{W^{1,2}} \leq C \left\| \psi - \int_{\mathbb{T}} \psi dy \right\|_{L^2} \leq C \|\psi\|_{W^{1,2}},$$

which proves $D_\varphi H(\bar{\Phi}_n, \cdot) \in C(\mathbb{R}, \mathcal{L}(W^{1,2}(\mathbb{T}), W^{1,2}(\mathbb{T})))$.

Finally, the analysis in Theorem 6.1 shows that $\ker(D_\varphi H(\bar{\Phi}_n, \kappa_n)) = \text{span}\{\varphi_n\}$ with $\varphi_n = \sqrt{2} \cos(2n\pi x)$, and $\varphi_n \notin \text{ran}(D_\varphi H(\bar{\Phi}_n, \kappa_n))$. Indeed,

$$\begin{aligned} G_\chi(0, 0)\varphi = 0 &\iff D_\varphi H(\bar{\Phi}_n, \kappa_n)\varphi = 0, \\ \varphi \in \text{ran}(G_\chi(0, 0)) &\iff (D\Delta - I)^{-1}\varphi \in \text{ran}(D_\varphi H(\bar{\Phi}_n, \kappa_n)), \end{aligned}$$

and thus zero is a simple eigenvalue of $D_\varphi H(\bar{\Phi}_n, \kappa_n)$.

Having verified all assumptions, the conclusion follows. $\square$

We now analyze the global behavior of the bifurcating branch in the subcritical regime.

Theorem 6.6 (Existence of turning point). *Suppose $\kappa_1 = 1 + D\mu_1 < 1.5$, corresponding to a subcritical pitchfork bifurcation at $(\bar{\Phi}_1, \kappa_1)$. Then there exists $\hat{s} \in \mathbb{R}$ such that the branch $\Gamma_1(s)$ exhibits a turning point at $(\bar{\Phi}(\hat{s}), \kappa_f)$, where $\kappa_f = \kappa_1(\hat{s}) \in (0, \kappa_1)$.*

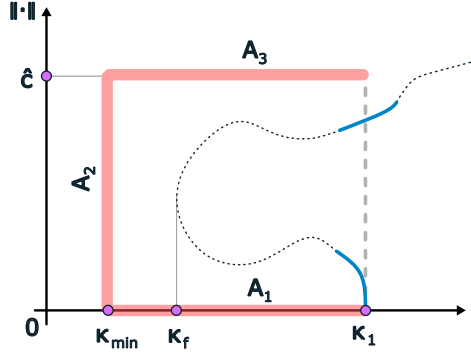


FIGURE 6.1.

Proof. We analyze the possible behavior of $\Gamma_1(s)$. By Theorem 6.5, the maximal connected component containing $\Gamma_1(s)$ either extends to infinity in $W^{1,2}(\mathbb{T}) \times \mathbb{R}$ or returns to the trivial branch at some $\kappa \neq \kappa_1$. Multiplying equation (2.2) by U and integrating yields

$$D \int_{\mathbb{T}} |U_x|^2 dx + \int_{\mathbb{T}} U^2 dx = \kappa \frac{\int_{\mathbb{T}} e^U U dx}{\int_{\mathbb{T}} e^U dy} \leq \kappa \|U\|_{L^\infty(\mathbb{T})}.$$

Therefore, by Sobolev embedding $\|U\|_{W^{1,2}}^2 \leq C\kappa \|U\|_{W^{1,2}}$, and hence all stationary solutions are uniformly bounded in $W^{1,2}$ for $\kappa \in [0, K]$. In particular, the branch $\Gamma_1(s)$ can not cross the horizontal line A_3 in Figure 6.1.

Since κ_1 is the smallest bifurcation point, the branch cannot intersect the trivial branch for $\kappa \in (0, \kappa_1)$; this corresponds to the horizontal line A_1 in Figure 6.1. Furthermore, Theorem 6.5 guarantees the uniqueness of the constant solution for $\kappa < \kappa_{\min}$, which forms the boundary A_2 .

Since the branch emerges from $(\bar{\Phi}_1, \kappa_1)$ with $\kappa'_1(0) = 0$ and $\kappa''_1(0) < 0$ (subcritical bifurcation), and remains confined within the bounded rectangle enclosed by A_1 , A_2 , and A_3 , Theorem 6.5 implies that the branch must exit this region. The only possible exit is through right boundary (gray dashed line).

By continuity there exists $\hat{s} \in \mathbb{R}$ which is a turning point of the branch $\Gamma_1(s)$. $\square$

Remark 6.7. We cannot analytically prove how the branch in $[\kappa_{\min}, \kappa_1]$ behaves in detail. Numerical simulations suggest that we always have the simplest case of one fold bifurcation at κ_f . At the fold bifurcation κ_f the unstable eigenvalue from the subcritical pitchfork bifurcation could get stable or one of the stable eigenvalues could lose stability. In the first case, the branch would be stable after the fold bifurcation and we would have the existence of stable nontrivial steady states for some $\kappa \in (\kappa_f, \kappa_1)$ and, thus, bistability as the constant steady state is also stable. This would not be necessarily true in the second case. However, in numerical analysis, we always observe a single fold bifurcation κ_f in $(0, \kappa_1)$ at which the branch $\Gamma(s)$ gets stable. Hence, we have bistability in the whole interval (κ_f, κ_1) .

Data availability statement: Data sharing is not applicable to this article as no datasets were generated or analyzed.

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APPENDIX A. MODEL DERIVATION

A.1. Hyper-elastic tissue model. We demonstrate the derivation of the mechanochemical patterning model for *Hydra* spheroids introduced in [41] in two spatial dimensions. It retains the essential coupling between mechanics and chemistry while simplifying the geometric structure. In this setting, the *Hydra* spheroid is represented as an elastic closed curve, providing a minimal framework to analyze the core mechanochemical feedback mechanisms, see Figure A.1.

The initial, stress-free reference configuration, denoted by $\mathcal{S}_0$, is assumed to be a circle of fixed circumference $L > 0$, parametrized by

$$\mathbf{g}(\xi) = \frac{L}{2\pi} \begin{pmatrix} \cos \frac{2\pi}{L}\xi \\ \sin \frac{2\pi}{L}\xi \end{pmatrix}, \quad \xi \in [0, L].$$

This reference curve is subjected to a spatial deformation $\mathbf{h} : \mathcal{S}_0 \rightarrow \mathcal{S}$, resulting in a new closed curve $\mathcal{S}$ with circumference $\ell > 0$. We prescribe a fixed enclosed area $A_0 > \text{Area}(\mathcal{S}_0)$ and restrict attention to deformations that satisfy $\text{Area}(\mathcal{S}) = A_0$. The deformed curve is parametrized by the composition

$$\mathbf{f} : [0, L] \rightarrow \mathcal{S}, \quad \mathbf{f} = \mathbf{h} \circ \mathbf{g}.$$

The deformation gradient in surface coordinates is given by the matrix

$$\mathbf{F} = \nabla_{\mathbf{X}} \mathbf{h}(\mathbf{X}) = \mathbf{f}'(\xi) \otimes \nabla(\mathbf{g}^{-1}(\mathbf{X})), \quad \mathbf{X} = \mathbf{g}(\xi).$$

Taking a derivative of the identity $\mathbf{g}^{-1}(\mathbf{g}(\xi)) = \xi$ and using $|\mathbf{g}'(\xi)| \equiv 1$ implies that $\nabla(\mathbf{g}^{-1}(\mathbf{X})) = \mathbf{g}'(\xi)^T$, where $\mathbf{g}'(\xi)$ (the covariant basis vector on $\mathcal{S}_0$) is a column vector and the contravariant basis vector $\mathbf{g}'(\xi)^T$ is a row vector. We therefore can write

$$\mathbf{F} = \mathbf{f}'(\xi) \otimes \mathbf{g}'(\xi)^T.$$

The *Right Cauchy-Green* deformation tensor reads

$$\mathbf{C} = \mathbf{F}^T \mathbf{F} = (\mathbf{g}'(\xi) \otimes \mathbf{f}'(\xi)^T) (\mathbf{f}'(\xi) \otimes \mathbf{g}'(\xi)^T) = |\mathbf{f}'(\xi)|^2 \mathbf{g}'(\xi) \otimes \mathbf{g}'(\xi)^T,$$

and the Green-Lagrangian strain is

$$(A.1) \quad \boldsymbol{\varepsilon} = \frac{1}{2}(\mathbf{C} - \mathbf{I}) = \frac{1}{2}(|\mathbf{f}'(\xi)|^2 - 1) \mathbf{g}'(\xi) \otimes \mathbf{g}'(\xi)^T,$$

where the identity in the tangent space of $\mathcal{S}_0$ is given by $\mathbf{I} = \mathbf{g}'(\xi) \otimes \mathbf{g}'(\xi)^T$.

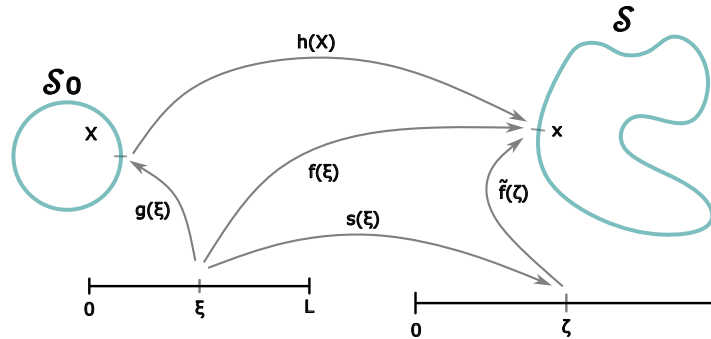


FIGURE A.1. Schematic representation of the elastic deformation framework for the *Hydra* spheroid model. Left: Reference configuration $\mathcal{S}_0$ parametrized by $\mathbf{g} : [0, L] \rightarrow \mathcal{S}_0$. Right: Deformed configuration $\mathcal{S}$ showing the composition of mappings $\mathbf{f} = \mathbf{h} \circ \mathbf{g}$ and the arc-length reparametrization $\tilde{\mathbf{f}}$ with $\mathbf{f} = \tilde{\mathbf{f}} \circ s$.

We model the tissue as a hyperelastic material governed by the Saint Venant-Kirchhoff constitutive law with strain energy density

$$W(\varepsilon) = \frac{\lambda}{2} \text{tr}^2(\varepsilon) + \frac{\mu}{2} \text{tr}(\varepsilon^2),$$

where the Lamé parameters satisfy $\lambda = 0$ and $\mu = E(u)$, with $u := u(\xi)$ the morphogen concentration and $E(u)$ the morphogen-dependent elastic modulus encoding the mechano-chemical coupling central to our model. The specific choice of parameters follows from the modelling considerations in [41]. The total elastic energy of a deformation is given by

$$\mathcal{E}_{\text{el}} = \int_{S_0} W(\varepsilon) dS_0 = \frac{1}{8} \int_0^L E(u) (|\mathbf{f}'(\xi)|^2 - 1)^2 |\mathbf{g}'(\xi)| d\xi,$$

where we used equation (A.1).

Assuming quasi-static mechanics due to fast tissue relaxation, the elastic energy is minimized under the area constraint

$$A_0 = A(f) = \frac{1}{2} \int_S \mathbf{x} \cdot \mathbf{n} dS = \frac{1}{2} \int_0^L \mathbf{f}(\xi) \cdot \mathbf{n}(\mathbf{f}(\xi)) |\mathbf{f}'| d\xi = -\frac{1}{2} \int_0^L \mathbf{f}(\xi) \cdot \mathbf{f}'(\xi)^\perp d\xi,$$

where $\mathbf{n}(\mathbf{x}) = -\mathbf{f}'(\mathbf{x})^\perp / |\mathbf{f}'(\mathbf{x})|$ is the outer unit normal vector on $\mathcal{S}$ at $\mathbf{x} \in \mathcal{S}$ and $^\perp$ represents the rotation of a 2-dimensional vector by $\pi/2$: $(a, b)^\perp = (-b, a)$.

To find minimizers of $\mathcal{E}_{\text{el}}$ we introduce Lagrange multipliers. The Gateaux derivative with respect to variations $\delta \mathbf{f}(\xi)$ with periodic boundary conditions yields

$$\begin{aligned} 0 &= D\mathcal{E}_{\text{el}}(\mathbf{f})(\delta \mathbf{f}) + p DA(f)(\delta f) \\ (A.2) \quad &= \frac{1}{2} \int_0^L E(u) (|\mathbf{f}'(\xi)|^2 - 1) \mathbf{f}'(\xi) \cdot \delta \mathbf{f}'(\xi) d\xi - p \int_0^L \delta \mathbf{f}(\xi) \cdot \mathbf{f}'(\xi)^\perp d\xi, \end{aligned}$$

where $p \in \mathbb{R}$, the inner pressure, acts as a Lagrange multiplier for the area constraint. Integrating by parts, we obtain that equation (A.2) is satisfied if and only if the minimizer f satisfies

$$\left(\sigma(\xi) \frac{\mathbf{f}'(\xi)}{|\mathbf{f}'(\xi)|} \right)' + p \mathbf{f}'(\xi)^\perp = 0,$$

where the stress is given by

$$\sigma(\xi) = E(u) \frac{1}{2} (|\mathbf{f}'(\xi)|^2 - 1) |\mathbf{f}'(\xi)|.$$

Evaluating the derivative we find that

$$(A.3) \quad \sigma'(\xi) \frac{\mathbf{f}'(\xi)}{|\mathbf{f}'(\xi)|} + \sigma(\xi) \left(\frac{\mathbf{f}'(\xi)}{|\mathbf{f}'(\xi)|} \right)' + p \mathbf{f}'(\xi)^\perp = 0.$$

Taking the scalar product of this with $\mathbf{f}'(\xi)$ and using that $\left(\frac{\mathbf{f}'(\xi)}{|\mathbf{f}'(\xi)|} \right)' \cdot \mathbf{f}'(\xi) = 0$ as well as the impenetrability assumption $|\mathbf{f}'(\xi)| > 0$ we obtain that $\sigma' \equiv 0$, i.e. the stress is constant. Taking the scalar product of (A.3) with $\mathbf{f}'(\xi)^\perp$ we obtain that $\sigma \left(\frac{\mathbf{f}'(\xi)}{|\mathbf{f}'(\xi)|} \right)' \cdot \mathbf{f}'(\xi)^\perp + p |\mathbf{f}'(\xi)|^2 = 0$, i.e. the curvature $\kappa = \frac{\mathbf{f}''(\xi)}{|\mathbf{f}'(\xi)|^3} \cdot \mathbf{f}'(\xi)^\perp = -p/\sigma$ (Laplace's Law) is constant along $\mathcal{S}$. Hence, the minimizer $\mathcal{S}$ must be circular with circumference $\ell > L$.

To separate geometric shape from tangential stretching, we introduce an arc-length parametrization:

$$\mathbf{f}(\xi) = \tilde{\mathbf{f}}(s(\xi)),$$

where $\tilde{\mathbf{f}} : [0, \ell] \rightarrow \mathcal{S}$ satisfies $|\tilde{\mathbf{f}}'| = 1$, and $s : [0, L] \rightarrow [0, \ell]$ with $s' > 0$ encodes the tangential deformation. Since $s'(\xi) = |f'(\xi)|$ represents the local stretch ratio (ratio of deformed to reference arc length $|g'(\xi)| \equiv 1$), the quantity

$$(A.4) \quad \tilde{\varepsilon} = \frac{1}{2}((s')^2 - 1)$$

is the finite Green-Lagrangian strain in one dimension.

Conservation of total length imposes

$$\int_0^L (s'(\xi) - 1) d\xi = \ell - L.$$

The morphogen concentration $u(t, \xi)$ evolves according to

$$\partial_t u = \kappa \tilde{\varepsilon} - \alpha u + D \partial_{\xi\xi} u,$$

with production proportional to local strain, linear degradation rate $\alpha > 0$, and diffusion coefficient $D > 0$. Taking together, the full mechanochemical system reads

$$\begin{cases} \partial_t u = \kappa \tilde{\varepsilon} - \alpha u + D \partial_{\xi\xi} u, \\ 0 = \partial_{\xi}(E(u)(s'^2 - 1)s'), \\ \ell - L = \int_0^L (s' - 1) d\xi, \end{cases} \quad \begin{matrix} \xi \in [0, L], \ t \geq 0, \\ \\ t \geq 0, \end{matrix}$$

with periodic boundary conditions and appropriate initial data. The 1D finite strain is defined by (A.4), completing the mechanochemical coupling.

A.2. Infinitesimal strain. To facilitate analytical treatment of the mechanochemical model, we invoke the infinitesimal strain approximation. This is justified biologically, as *Hydra* cells experience only small-amplitude distortion during mechanochemical signalling, so that the local deformation gradients remain close to the identity [41]. Under the assumption that $|s' - 1| \ll 1$, the finite strain $\tilde{\varepsilon} = \frac{1}{2}(s'^2 - 1)$ can be replaced by its linearized counterpart

$$\hat{\varepsilon}(t, \xi) := s'(t, \xi) - 1 \approx \tilde{\varepsilon}(t, \xi),$$

reducing the nonlinear system to a tractable linear framework while retaining the essential mechanochemical coupling.

Following the minimization procedure, the mechanical equilibrium condition reduces under the infinitesimal strain approximation to

$$0 = \frac{1}{2} \partial_{\xi}(E(u)(s'^2 - 1)s') = \frac{1}{2} \partial_{\xi}(E(u)(2(s' - 1) + (s' - 1)^2)s') \approx \partial_{\xi}(E(u)\hat{\varepsilon}),$$

where $\hat{\varepsilon} := s' - 1$ is the linearized strain. Consequently, the full mechanochemical system in the infinitesimal strain approximation, with $\varepsilon := \hat{\varepsilon}(\xi, t)$, reads

$$\begin{cases} \partial_t u = \kappa \varepsilon - \alpha u + D \partial_{\xi\xi} u, \\ 0 = \partial_{\xi}(E(u)\varepsilon), \\ \ell - L = \int_0^L \varepsilon d\xi, \end{cases} \quad \begin{matrix} \xi \in [0, L], \ t \geq 0, \\ \\ t \geq 0. \end{matrix}$$

APPENDIX B. WELL-POSEDNESS OF THE MODEL

We provide here a proof of existence and uniqueness of a global-in-time solution to problem (2.1). A similar approach can be found in [28]; we include the details here for the sake of completeness.

Proof of proposition 2.1. We begin by establishing the existence of a local-in-time solution. Let $u_0 \in W^{1,2}(\mathbb{T})$ and consider the mild formulation of problem (2.1):

$$u(t, \cdot) = e^{D\Delta t} u_0 + \int_0^t e^{D\Delta(t-s)} \left(\kappa \frac{e^{u(s, \cdot)}}{\int_0^1 e^{u(s, y)} dy} - u(s, \cdot) \right) ds.$$

Here, $e^{D\Delta t}$ is a semigroup generated on $L^2(\mathbb{T})$ by the second derivative $D\Delta$ with domain $D(D\Delta) = W^{2,2}(\mathbb{T})$. Let $X = C([0, \tau]; W^{1,2}(\mathbb{T}))$ be the Banach space with the norm $\|v\|_X = \sup_{0 \leq t \leq \tau} \|v(\cdot, t)\|_{W^{1,2}}$ where τ is a positive constant to be determined further below. We define the map $F : X \rightarrow X$ by

$$F(v) := e^{D\Delta t} u_0 + \int_0^t e^{D\Delta(t-s)} \left(\kappa \frac{e^{v(s, \cdot)}}{\int_0^1 e^{v(s, y)} dy} - v(s, \cdot) \right) ds.$$

We further introduce the ball

$$V = \{v \in X : \|v\|_X \leq 3\delta\} \quad \text{where} \quad \delta = \|u_0\|_{W^{1,2}}.$$

Notice that by Sobolev embedding, any $v \in V$ satisfies $\|v(t, \cdot)\|_{L^\infty} \leq C\delta$ for a constant $C > 0$. It is well known that the semigroup $e^{D\Delta t}$ satisfies the estimates

$$\|e^{D\Delta t} v\|_{L^2} \leq C\|v\|_{L^2} \quad \text{and} \quad \|\nabla e^{D\Delta t} v\|_{L^2} \leq C t^{-\frac{1}{2}} \|v\|_{L^2} \quad \text{for } v \in L^2(\mathbb{T}).$$

These estimates follow from the explicit representation of $e^{D\Delta t}$ as the kernel is given by $\mathbb{Z}$ -periodization of the heat kernel satisfying the contractivity estimates [33]. Consequently, for $0 \leq t \leq \tau$, we derive the estimate

$$\begin{aligned} \|F(v)\|_{W^{1,2}} &\leq \|e^{D\Delta t} u_0\|_{W^{1,2}} + C \int_0^t \left\| e^{D\Delta(t-s)} \left(\kappa \frac{e^{v(s, \cdot)}}{\int_0^1 e^{v(s, y)} dy} - v(s, \cdot) \right) \right\|_{W^{1,2}} ds \\ &\leq \|e^{D\Delta t} u_0\|_{W^{1,2}} + C \int_0^t \left(1 + (t-s)^{-\frac{1}{2}} \right) \left(\kappa \frac{\|e^{v(s, \cdot)}\|_{L^2}}{\|e^{v(s, \cdot)}\|_{L^1}} + \|v(s, \cdot)\|_{L^2} \right) ds. \end{aligned}$$

Since v is uniformly bounded, we have $\|e^v\|_{L^2} \leq e^{\|v\|_{L^\infty}}$ and $\|e^v\|_{L^1} \geq e^{-\|v\|_{L^\infty}}$. Thus, we obtain

$$\begin{aligned} \|F(v)\|_{W^{1,2}} &\leq \|e^{D\Delta t} u_0\|_{W^{1,2}} + C \int_0^t \left(1 + (t-s)^{-\frac{1}{2}} \right) \left(\kappa \frac{e^{\|v(s, \cdot)\|_{L^\infty}}}{e^{-\|v(s, \cdot)\|_{L^\infty}}} + \|v(s, \cdot)\|_2 \right) ds \\ &\leq \|e^{D\Delta t} u_0\|_{W^{1,2}} + C(\kappa e^{2C\delta} + 3\delta) (t + \sqrt{t}) \end{aligned}$$

using $v \in V$ and that the semigroup $e^{D\Delta t}$ is bounded on $W^{1,2}(\mathbb{T})$. By choosing $\tau > 0$ sufficiently small, we obtain the estimate $\|F(v)\|_X \leq 3\delta$.

Next we show that F is a contraction mapping on the ball V . We have

$$\|F(v) - F(w)\|_{W^{1,2}} \leq \int_0^t \left(1 + (t-s)^{-\frac{1}{2}} \right) \left\| \kappa \frac{e^v}{\int_0^1 e^v dy} - \kappa \frac{e^w}{\int_0^1 e^w dy} - v + w \right\|_{L^2} ds.$$

Observing that $e^v / \int e^v - e^w / \int e^w = (e^v - e^w) / \int e^v + e^w (\int e^w - \int e^v) / (\int e^v \int e^w)$ and letting $C(t) = t + \sqrt{t}$, we obtain

$$\|F(v) - F(w)\|_{W^{1,2}} \leq C(t) \|v - w\|_X + C(t) \kappa \sup_{t \in [0, \tau]} \left\| \kappa \frac{e^v - e^w}{\int_0^1 e^v dy} + \kappa e^w \frac{\int_0^1 e^w dy - \int_0^1 e^v dy}{\int_0^1 e^v dy \int_0^1 e^w dy} \right\|_{L^2}.$$

Using L^∞ -estimates for v and w we derive

$$e^w \left| \int_0^1 (e^w - e^v) dy \right| / \left(\int_0^1 e^v dy \int_0^1 e^w dy \right) \leq e^{3C\delta} \int_0^1 |e^v - e^w| dy$$

and

$$|e^v - e^w| / \int_0^1 e^v dy \leq e^{C\delta} |e^v - e^w|.$$

Thus, we obtain

$$\|F(v) - F(w)\|_{W^{1,2}} \leq C(t) \|v - w\|_X + C(t) \kappa e^{C\delta} \sup_{t \in [0, \tau]} \|e^v - e^w\|_{L^2} + C(t) \kappa e^{3C\delta} \sup_{t \in [0, \tau]} \|e^v - e^w\|_{L^\infty}.$$

By the Sobolev embedding and mean value argument, we conclude

$$\|F(v) - F(w)\|_{W^{1,2}} \leq C(t) \|v - w\|_X + C(t) \sup_{t \in [0, \tau]} \|v - w\|_{W^{1,2}} \leq C(\tau) \|v - w\|_X.$$

By choosing τ sufficiently small, the proof is completed by the Banach fixed point argument as $C(\tau) \downarrow 0$ as $\tau \downarrow 0$.

Next, we demonstrate that the solution can be extended globally in time. The Lyapunov function defined in (4.1) satisfies

$$\frac{d}{dt} \mathcal{J}(u(\cdot, t)) = D\mathcal{J}(u(\cdot, t)) u_t(\cdot, t) = -\|u_t(\cdot, t)\|_{L^2}^2 \leq 0,$$

which implies that $\mathcal{J}(u(\cdot, t)) \leq \mathcal{J}(u_0)$. Using Sobolev embedding, we obtain

$$\begin{aligned} \frac{D}{2} \int_0^1 |u_x(x, t)|^2 dx + \frac{1}{2} \int_0^1 |u(x, t)|^2 dx &\leq \mathcal{J}(u(\cdot, t)) + \kappa \log \left(\int_0^1 e^{u(x, t)} dx \right) \\ &\leq \mathcal{J}(u_0) + \kappa C \|u(\cdot, t)\|_{W^{1,2}}. \end{aligned}$$

This yields the inequality $\|u(\cdot, t)\|_{W^{1,2}}^2 \leq C + \|u(\cdot, t)\|_{W^{1,2}}$ which implies $\|u(\cdot, t)\|_{W^{1,2}} < C(u_0)$ for all $t \geq \tau$. Similarly, we derive

$$\int_0^t \|u_t(\cdot, t)\|_{L^2}^2 = J(u_0) - J(u(\cdot, t)) \leq J(u_0) + C \|u(\cdot, t)\|_{W^{1,2}} + \kappa C \|u(\cdot, t)\|_{L^\infty} \leq C,$$

where the bound is uniform with respect to t . Furthermore, we have

$$\begin{aligned} \int_0^t \|\Delta u(\cdot, t)\|_{L^2}^2 &= \int_0^t \left\| u_t(\cdot, t) + u(\cdot, t) + \kappa \frac{e^{u(\cdot, t)}}{\int_0^1 e^{u(y, t)} dy} \right\|_{L^2}^2 \\ &\leq \int_0^t \|u_t(\cdot, t)\|_{L^2}^2 + \int_0^t \|u(\cdot, t)\|_{L^2}^2 + \int_0^t \left\| \kappa \frac{e^{u(\cdot, t)}}{\int_0^1 e^{u(y, t)} dy} \right\|_{L^2}^2 \leq C + Ct. \end{aligned}$$

These estimates ensures that the solution exists for all time $t \geq 0$ and satisfies

$$u \in C([0, \infty), W^{1,2}(\mathbb{T})), \quad u \in L^2([0, \infty), L^2(\mathbb{T})), \quad u \in L^2((0, T), W^{2,2}(\mathbb{T})),$$

for each $T > 0$. To establish uniqueness, we employ the theory of dynamical systems. Since the solutions are bounded, it follows from [16, Thm 3.4.1] that the dependence of solutions of initial conditions is continuous, thereby defining dynamical system. Consequently the solutions are unique. $\square$

APPENDIX C. COMPUTATIONS OF THE DERIVATIVES

Define the mapping $G : X \times \mathbb{R} \rightarrow Y$ by

$$G(\chi, \alpha) := D\chi_{xx} - \chi + (\kappa_n + \alpha) \left(\frac{e^x}{\int e^x dy} - 1 \right),$$

where we use the notation of Section 6. The derivatives of G with respect to $\chi \in X$ and $\alpha \in \mathbb{R}$ are:

$$\begin{aligned} G_\chi(\chi, \alpha)(h) &= Dh_{xx} - h + (\kappa_n + \alpha) \left[\frac{e^x}{\int e^x dy} h - \frac{e^x}{(\int e^x dy)^2} \int e^x h dy \right], \\ G_{\chi\chi}(\chi, \alpha)(h_1, h_2) &= (\kappa_n + \alpha) \left[\frac{e^x}{\int e^x dy} h_1 h_2 + 2 \frac{e^x}{(\int e^x dy)^3} \int e^x h_1 dy \int e^x h_2 dy \right. \\ &\quad \left. - \frac{e^x}{(\int e^x dy)^2} \left(h_1 \int e^x h_2 dy + h_2 \int e^x h_1 dy + \int e^x h_1 h_2 dy \right) \right], \\ G_{\chi\chi\chi}(\chi, \alpha)(h, h, h) &= (\kappa_n + \alpha) \left[\frac{e^x}{\int e^x dy} h^3 - \frac{e^x}{(\int e^x dy)^2} \left(3h^2 \int e^x h dy + 3h \int e^x h^2 dy + \int e^x h^3 dy \right) \right. \\ &\quad \left. + \frac{e^x}{(\int e^x dy)^3} \left(8 \int e^x h dy \int e^x h^2 dy + 6h \left(\int e^x h dy \right)^2 + 2 \int e^x h dy \int e^x h^2 dy \right) \right. \\ &\quad \left. - 6 \frac{e^x}{\int e^x dy} \left(\int e^x h dy \right)^3 \right], \\ G_\alpha(\chi, \alpha) &= \left(\frac{e^x}{\int e^x dy} - 1 \right), \\ G_{\chi\alpha}(\chi, \alpha)(h) &= \frac{e^x}{\int e^x dy} h - \frac{e^x}{(\int e^x dy)^2} \int e^x h dy \end{aligned}$$

for $h, h_1, h_2 \in X$ and $\beta \in \mathbb{R}$.